

Report of AAPM Task Group 155: Megavoltage photon beam dosimetry in small fields and non-equilibrium conditions

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Abstract

Small-field dosimetry used in advance treatment technologies poses challenges due to loss of lateral charged particle equilibrium (LCPE), occlusion of the primary photon source, and the limited choice of suitable radiation detectors. These challenges greatly influence dosimetric accuracy. Many high-profile radiation incidents have demonstrated a poor understanding of appropriate methodology for small-field dosimetry. These incidents are a cause for concern because the use of small fields in various specialized radiation treatment techniques continues to grow rapidly. Reference and relative dosimetry in small and composite fields are the subject of the International Atomic Energy Agency (IAEA) dosimetry code of practice that has been published as TRS-483 and an AAPM

Abbreviations: %dd(10)_x, The photon component of the percent depth dose at 10 cm depth in water for a 10 cm × 10 cm field; ($\bar{\mu}_{en}/\rho$), Spectrum-averaged mass energy-absorption coefficient; AAPM, American Association of Physicists in Medicine; CPE, Charged Particle Equilibrium; DLG, Dosimetric leaf gap; $D_{w,Q_{ref}}^{ref}$, Absorbed dose to water at the reference depth z_{ref} in water in the absence of the detector in a field specified by f_{msr} and beam quality Q_{msr} ; f_{clin} , Clinical (clin) non-reference radiation field; f_{msr} , Machine-specific reference (msr) field; f_{ref} , Reference field (ref) specified in dosimetry protocols for which the calibration coefficient of an ionization chamber in terms of absorbed dose to water is provided by a standards laboratory; FWHM, Full-width at half-maximum; GUM, Guide to the expression of Uncertainty in Measurements; IAEA, International Atomic Energy Agency; ICRU, International Commission on Radiation Units and Measurements; IMRT, Intensity-modulated radiation therapy; k_{Q,Q_0} , The beam-quality correction factor, which corrects for the differences between the response of an ionization chamber in the reference beam of quality Q_0 used for calibrating the chamber and the beam of quality Q (defined as k_{Q_0} in TG-51⁴); k_{Q,Q_0}^{ref} , Correction factor that accounts for the differences between the response of a detector in field f_{ref} in a beam of quality Q and reference beam quality Q_0 as defined in TRS-483¹ and Palmans et al²; $k_{Q_{clin},Q_{msr}}^{ref}$, The detector-specific correction factor that accounts for the difference between the responses of the detector in fields f_{clin} in a beam of quality Q_{clin} and in fields f_{msr} in beam of quality Q_{msr} as defined by Alfonso et al.³; LCPE, Lateral charged particle equilibrium; $M_{Q_{msr}}^{msr}$, Detector reading in field f_{msr} and beam quality Q_{msr} corrected for influence of changes in pressure and temperature, incomplete charge collection, polarity effect and electrometer correction factor (TRS-483¹); MU, Monitor unit; N_{D,w,Q_0} , This is $N_{D,w}$ in TG-51⁴ and defined as the calibration coefficient in terms of absorbed dose to water for an ionization chamber at a reference beam of quality Q_0 and field size f_{ref} ; N_{D,w,Q_0}^{ref} , Calibration coefficient in terms of absorbed dose to water for an ionization chamber at a reference beam quality Q_0 in the conventional reference field f_{ref} ; OAR, Off-Axis Ratio; PDD, Percent Depth Dose; P_{eff} , Effective point of measurement (EPOM); pf_{clin} , Detector perturbation correction factor in clinical field, f_{clin} ; pf_{msr} , Detector perturbation factor in machine-specific reference field, f_{msr} ; Q_0 , The reference beam quality in a standards lab, usually ⁶⁰Co; $Q_{clin,Q_{msr}}^{ref}$, Field Output factor that converts the absorbed dose to water for the machine-specific reference field f_{msr} to the absorbed dose to water for the clinical field f_{clin} ; Q_{clin} , The beam quality of the clinical non-reference radiation field, f_{clin} ; Q_{msr} , The beam quality of the machine-specific reference field, f_{msr} ; Q_{ref} , The beam quality of the conventional reference field, f_{ref} ; r_{LCPE} , The minimum photon beam radius required to achieve LCPE; SBRT, Stereotactic body radiation therapy; S_{clin} , Equivalent field size defined in terms of FWHM; SFD, Stereotactic Photon Diode; SRS, Stereotactic radiosurgery; SSD, Source-to-Surface Distance the term is introduced for machines that cannot establish the conventional reference field (i.e., 10 cm × 10 cm). For example, f_{msr} in Gamma Knife, CyberKnife and TomoTherapy are 1.6 or 1.8 cm diameter, 6 cm diameter and 5 cm × 10 cm, respectively. Some standards laboratories provide calibration of ionization chambers in the hospitals msr field. This is not widely available yet; TMR, Tissue Maximum Ratio; TPR, Tissue Phantom Ratio; TPR_{10}^{20} , The tissue phantom ratio in water at depth of 20 and 10 cm for a field size of 10 cm × 10 cm defined at a source to detector distance of 100 cm; VMAT, Volumetric modulated arc radiotherapy.

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summary publication (IAEA TRS 483; Dosimetry of small static fields used in external beam radiotherapy: An IAEA/AAPM International Code of Practice for reference and relative dose determination, Technical Report Series No. 483; Palmans et al., Med Phys 45(11):e1123, 2018). The charge of AAPM task group 155 (TG-155) is to summarize current knowledge on small-field dosimetry and to provide recommendations of best practices for relative dose determination in small megavoltage photon beams. An overview of the issue of LCPE and the changes in photon beam perturbations with decreasing field size is provided. Recommendations are included on appropriate detector systems and measurement methodologies. Existing published data on dosimetric parameters in small photon fields (e.g., percentage depth dose, tissue phantom ratio/tissue maximum ratio, off-axis ratios, and field output factors) together with the necessary perturbation corrections for various detectors are reviewed. A discussion on errors and an uncertainty analysis in measurements is provided. The design of beam models in treatment planning systems to simulate small fields necessitates special attention on the influence of the primary beam source and collimating devices in the computation of energy fluence and dose. The general requirements for fluence and dose calculation engines suitable for modeling dose in small fields are reviewed. Implementations in commercial treatment planning systems vary widely, and the aims of this report are to provide insight for the medical physicist and guidance to developers of beams models for radiotherapy treatment planning systems.

KEYWORDS
detector correction factors, non-equilibrium dosimetry, photon dose modeling, relative dosimetry, small field, treatment modalities

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1 | INTRODUCTION

The use of small photon fields has grown rapidly with the implementation of modern specialized radiation treatment techniques and dosimetry has been discussed in international forum as described in several publications.¹⁻³ The dosimetry and commissioning of traditional field sizes ranging from 4 cm × 4 cm to 40 cm × 40 cm typically used in radiation therapy has been addressed in several task groups.^{4,5} Documents providing guidance and the implementation of stereotactic radiotherapy (SRT), stereotactic radiosurgery (SRS),^{6,7} and stereotactic body therapy (SBRT)⁸ have been published with limited coverage of small-field dosimetry. The International Commission on Radiation Units and Measurements (ICRU) published Report 91⁹ on prescribing, recording, and reporting of stereotactic treatments with small photon beams which covers principles of small-field dosimetry and small-field treatment planning dose calculation algorithms. In linear accelerator-based SRS/SRT/SBRT, field sizes are defined by secondary/tertiary collimators (multileaf collimators (MLCs) and cones). Cones are divergent and are used to define circular fields with diameters typically varying from 0.5 cm to 4 cm.¹⁰ Detailed information on SRS/SRT dedicated systems can be found in

various references.^{6,7,11} Table 1 summarizes the smallest collimator settings available on such radiotherapy delivery systems.

In intensity-modulated radiotherapy (IMRT), the intensity of a radiation beam is varied either through multiple static MLC segments (step-and-shoot technique) or MLC leaves moving continuously as radiation is delivered (sliding window technique).^{12,13} Volumetric modulated arc therapy (VMAT) technology¹⁴⁻¹⁶ is used to more efficiently deliver highly conformal dose distributions as an alternative to sliding window and step-and-shoot IMRT.¹⁷ These techniques invariably use combinations of a large number of small fields. Dosimetric aspects specific to sliding window IMRT,¹⁴ the physical and dosimetric aspects of a dynamic MLC, and the variation on dose output have been extensively studied.^{18,19} The task groups TG-135 and TG-120 provide detailed discussion on the dosimetric tools and techniques for SRS and IMRT.^{11,20}

In the context of small fields, lateral charged particle equilibrium (LCPE)¹ (earlier referred as lateral electronic equilibrium, LEE²¹) is not established in the region of a radiation detector. The loss of LCPE indicates that electrons traveling laterally out of the region are not replaced by electrons moving laterally into it.²²⁻²⁴ The loss of LCPE occurs in the penumbra region, the interior portion of narrow fields, and regions with media of different densities (low density and interfaces).²⁴⁻²⁷ The sensitive volume of a detector, its density, and atomic composition can strongly influence dosimetric accuracy when LCPE is lost. In very narrow fields (such as 1 cm × 5 cm), LCPE is not established even on the central axis; consequently, small-field conditions exist in these fields even when using detectors with very small sensitive volumes and width of 1–2 mm. Under small-field conditions, the degree of LCPE on-axis and the output factor decrease rapidly with decreasing field size.^{28,29} Beam output is reduced further as the beam collimating system starts to occlude the

TABLE 1 Reference and minimum collimator settings available on radiotherapy treatment units delivering small photon fields

Treatment unit	Delivery technique	Minimum field size	Machine-specific reference (<i>msr</i>) field (cm ²)	Relevant task group report
Linac	IMRT/VMAT	0.1 × 0.1 cm ²	10 × 10	Ref. [20, 302]
Linac	SRS	0.5-cm ² diameter cone	10 × 10	Ref. [6, 10, 329]
Linac ^a	SBRT	0.1 × 0.5 cm ²	10 × 10	TG-101 [8, 330]
Tomotherapy	IMRT	1 × 40 cm	5 × 40	TG-148 [331]
CyberKnife (CK) ^b	IMRT	0.5-cm diameter	6.0-cm diameter	TG-135[11]
Gamma Knife	SRS	0.4-cm diameter	1.6- or 1.8-cm diameter cone	Ref. [6, 7]

Note: See main text for the definition of machine-specific reference field.

^aWithin their planning and delivery systems, each institution is able to administratively control their preferred smallest field setting for commissioning of the MLC/microMLC for SRS/SBRT techniques.

^bFor CK, the current minimum clinically used cone diameter is 4 mm (@ 65-cm SAD for trigeminals). The new generation of MLC-shaped CK machines includes the capability of 10 cm × 10 cm field size as noted in the Canadian report on CK by Vandervoort et al.³³²

primary radiation source.^{30,31} Electrons travel greater distances in low-density media such as lung and air, causing LCPE to break down and further complicating dosimetry.^{32,33}

Detectors of appropriate size, composition, and construction are needed for accurate dosimetric measurements in small fields. Many detectors are too large and cannot resolve penumbra widths of the beam profiles. Detector elemental composition impacts response in small fields to a lesser extent than the density.^{23,27,34-37} Numerous experimental and Monte Carlo (MC) studies have investigated the suitability of detectors for the determination of dosimetric parameters, such as percentage depth dose (PDD), tissue maximum and phantom ratio (TMR, TPR), and field output factors (Ω), in small photon fields.^{3,21,36,38-50} The consensus has been that suitable detectors must be considerably smaller than the field size as it impacts detector readings when CPE breaks down around the sensitive volumes of detectors.^{3,51,52}

Studies from the early 2000s indicated the existence of substantial problems with the determination of absorbed dose and machine output in small fields. In the wake of prominent cases of misadministration of radiation therapy in Europe and North America, several organizations commissioned working groups to systematically address small-field dosimetry. These included the British Society, Institute of Physics and Engineering in Medicine (IPEM), the AAPM, the International Atomic Energy Agency (IAEA), the International Commission of Radiation Units and Measurements (ICRU), as well as other national organizations around the world. The IPEM summarized the challenges associated with the dosimetry of small fields and provided an early extensive review on detectors and their merits in small fields in IPEM Report 103.³⁰ The IPEM Report also included recommendations on good practice in small-field dosimetry. Around 2007, within the AAPM two initiatives emerged in parallel. One initiative, in collaboration with the International Atomic Energy Agency (IAEA) and termed here the IAEA-AAPM initiative, focused on the aspect of reference dosimetry in linac beams that could not realize the standard TG-51 conditions as well as small-field linac output. A parallel initiative emerged in 2007 was approved as TG-155 and focused on the details of small-field relative dose measurements and small-field aspects impacting treatment planning dose calculations.

The IAEA-AAPM initiative led to a formalism, commonly known as the “Alfonso formalism,” for reference dosimetry of small and nonstandard radiation fields.³ Based on this formalism, IAEA-AAPM report TRS-483¹ (henceforth referred to as TRS-483) was developed that addresses calibration in nonstandard reference fields and linac output in static small fields. TRS-483 provides a code of practice for reference dosimetry for two specific sets of devices for which the conventional

reference field conditions such as a 10 cm × 10 cm field size and 100 cm source-to-surface distance (SSD) or source-to-axis distance (SAD) are not applicable: (a) the specialized treatment devices such as Gamma Knife, TomoTherapy, and CyberKnife and (b) linear accelerators with small fields with configurations that do not satisfy the reference calibration conditions defined in existing dosimetry protocols.^{4,53} TRS-483 provides calibration guidelines for absorbed dose in water traceable to a primary standard laboratory using a reference ion chamber. The report introduces the concept of the machine-specific reference (*msr*) field as many of the systems are unable to achieve the reference field size of 10 cm × 10 cm. For the determination of output in small static fields, the TRS-483 also presents compiled field output correction factors for detectors based on published literature at the time of publication.

While advanced guidance was being developed for small-field dosimetry, there was an increase in the availability of machines with flattening filter-free (FFF) beams on the latest design c-arm or robotic linacs which are being used to deliver SRS, SBRT, IMRT, and VMAT.⁵⁴ Lechner et al⁵⁵ have investigated small-field dosimetry in such beams (FFF). The TRS-483¹ report and other guidance documents^{56,57} provide physicists with information on the impact on measurements of the very high dose rates and non-flat profiles in FFF beams. Additionally, very small beams with millimeter widths are being used for animal irradiation but mainly with kilovoltage beams (e.g., 220 kVp with Cu filter).⁵⁸ Their dosimetry is beyond the scope of this report and will not be addressed here.

The purpose of the present TG-155 report is to describe the framework for applying detector-specific correction factors and the practical aspects of relative dosimetry of small fields. The report provides guidance and recommendations on the various types of active and passive detectors as well as guidance on measuring beam profiles, PDD, TMR/TPR in small radiation fields for beams from accelerating potential of 6 MV or lower since they are most often employed for specialized treatment procedures utilizing small fields. ICRU-91⁹ contains a section on small-field dosimetry and treatment planning systems (TPSs) without referencing individual manufacturers. For several reasons such as the greater range of secondary electrons, possible limitations in the accuracy of dose calculation algorithms, as well as neutron production, ICRU-91 recommends not using energies above 10 MV for small fields.⁹ Sources of errors in dosimetric measurements of small fields are analyzed, and estimates are provided of expected uncertainties with 1 σ in measured beam profiles, PDDs, TMRs, and field output factors.

The input of data in a TPS measured with an inappropriately chosen detector has been one of the root causes of significant dosimetric errors which have resulted in patient harm reported in the media.⁵⁹⁻⁶² Solberg

et al.⁶³ provided an executive summary of these major incidences in SRS and SBRT. Care has been taken to ensure that the report TG-155 is complementary and consistent with several important international reports while minimizing overlap of information. In this task group report, we provide a general overview of photon beam modeling used in dose calculation algorithms implemented in TPSs and in independent monitor unit (MU) verification software tools. Emphasis is placed on treatment planning dose calculation modeling in the case of narrow radiation fields. Many TPSs were originally designed and commissioned to describe fields larger than 3x3 cm², and an understanding of beam modeling is important when commissioning those systems to extend their use to small static or composite fields.⁶⁴ Additionally, key algorithm parameters influencing doses calculated in small fields are highlighted, and the importance of well-measured dosimetric small-field data is emphasized. This report also emphasizes the importance and methodology to accurately measure dosimetric data used for the configuration of dose calculation models.

2 | BRIEF OUTLINE OF IAEA/ AAPM FORMALISM

2.1 | Reference dosimetry

The IAEA and the AAPM jointly published a new code of practice (CoP) for reference and relative dosimetry of small static photon fields used in external beam radiotherapy published as IAEA Technical Report Series TRS-483¹ and summarized in a paper by Palmans et al.² Although Alfonso et al.³ provided options for

calibration of static field as well as composite fields like IMRT, only static field recommendations are made in the TRS-483¹ protocol and Palmans et al.² The formalism for reference dosimetry in small static photon fields is a slightly modified version of the formalism recommended by Alfonso et al.³ and essentially provides an extension to existing protocols for reference dosimetry.^{4,53} It is based on the use of an ionization chamber that has been calibrated by a standards laboratory in a reference beam of quality Q_0 . It defines the formalism and the necessary factors to account for possible changes in detector response to determine the absorbed dose to water under non-standard reference conditions, where the reference field size of 10 cm × 10 cm cannot be realized, as illustrated in Figure 1. According to the formalism, the absorbed dose to water at the reference depth in the absence of the detector, $D_{w,Q_{msr}}^{f_{msr}}$ in a field size f_{msr} of beam quality Q_{msr} is given by:

$$D_{w,Q_{msr}}^{f_{msr}} = M_{Q_{msr}}^{f_{msr}} N_{D,w,Q_0}^{f_{ref}} k_{Q,Q_0}^{f_{msr},f_{ref}} k_{Q_{msr},Q_0}^{f_{msr},f_{ref}} \quad (1a)$$

$$D_{w,Q_{msr}}^{f_{msr}} = M_{Q_{msr}}^{f_{msr}} N_{D,w,Q_0}^{f_{ref}} k_{Q,Q_0}^{f_{ref}} k_{Q_{msr},Q_0}^{f_{msr},f_{ref}} \quad (1b)$$

where subscripts and superscripts are defined in the section with the list of symbols. Briefly, the notation “msr” stands for the machine specific reference as shown in Table 1. M is the measurement reading by the detector (corrected for variations in ambient conditions, polarity, leakage, stem, and recombination effects). The notation f_{ref} denotes the conventional reference field in dosimetry protocols. The calibration coefficient

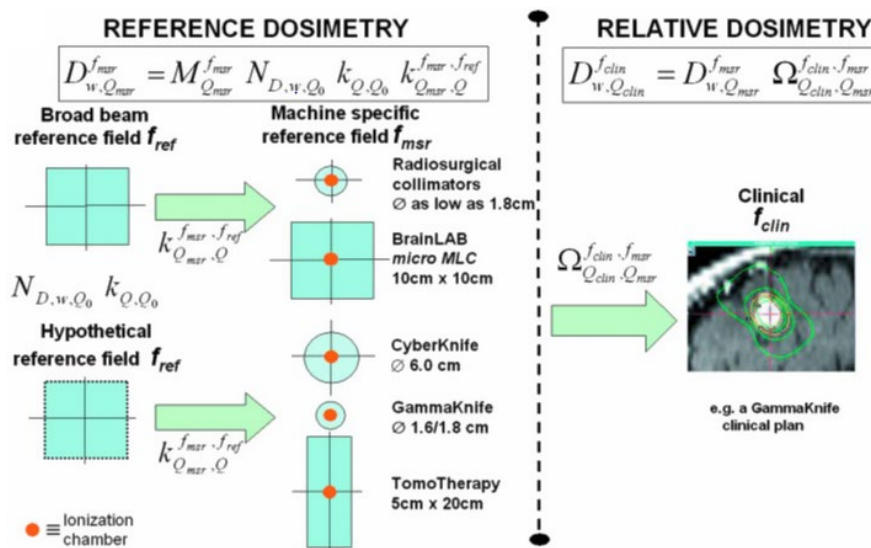


FIGURE 1 Schematic overview of the dosimetry of small static fields with reference field, according to the formalism proposed by Alfonso et al.³ The flowchart indicates the process in achieving dose in f_{msr} . The figure was adapted from Alfonso et al.³ for the static field calibration process [Color figure can be viewed at wileyonlinelibrary.com]

of an ionization chamber in terms of absorbed dose to water is provided by an Accredited Dosimetry Calibration Laboratory (ADCL) for the beam quality Q_0 for a specific field size f_{ref} . The f_{msr} is the machine-specific reference field, N_{D,w,Q_0} is the chamber-specific calibration coefficient in terms of absorbed dose to water for ^{60}Co in f_{ref} and k_{D,w,Q_0}^{ref} is the chamber-specific beam-quality correction factor. The last factor in Equation 1, $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ is a correction factor that accounts for the detector's difference in calibration coefficient between the conventional reference field and the actual f_{msr} field due to differences in field geometry, phantom material, and beam quality. The Equation 1(b) is used when no generic correction factor for the msr is available. The term $k_{Q_0,Q_0}^{f_{ref}}$ represents the difference between the response of the ionization chamber between beam quality Q and Q_0 at a standard laboratory.²

2.2 | Relative dosimetry

Relative dosimetry typically refers to the determination of ratio of absorbed doses which may either be in reference or non-reference conditions. Dosimetric quantities such as PDD, TMR, OAR, and field output factor are, by definition, ratios of doses and fall under relative dosimetry. Following the IAEA-AAPM task group's formalism, the ratio of detector signal under non-reference conditions can be related to the ratio of absorbed doses through a detector-specific correction factor. This factor accounts for the difference in the detector response between the arbitrary, non-reference, irradiation geometry and the reference irradiation geometry. Thus, absorbed dose to water at a point in a phantom in the absence of the detector for an arbitrary clinical field f_{clin} of beam quality Q_{clin} is expressed as:

$$D_{W,Q_{clin}}^{f_{clin}} = D_{W,Q_{msr}}^{f_{msr}} \Omega_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}} \quad (2)$$

where $\Omega_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ is the field output factor that converts the absorbed dose to water per monitor unit (MU) for the machine-specific reference field to the absorbed dose to water in the clinical field per MU.¹ This in essence is the traditional output factor or the in-water output ratio in TG-74⁵² but it is emphasized that the field output factor is a ratio of absorbed doses and not a ratio of detector readings. This is made explicit by writing the field output factor as

$$\Omega_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}} = \frac{D_{W,Q_{clin}}^{f_{clin}}}{D_{W,Q_{msr}}^{f_{msr}}} = \left[\frac{M_{Q_{clin}}^{f_{clin}}}{M_{Q_{msr}}^{f_{msr}}} \right] k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}} \quad (3)$$

and recognizing that a field output *correction* factor given by,

$$k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}} = \frac{D_{W,Q_{clin}}^{f_{clin}}}{D_{W,Q_{msr}}^{f_{msr}}} \frac{M_{Q_{msr}}^{f_{msr}}}{M_{Q_{clin}}^{f_{clin}}} \quad (4)$$

must be involved in determining the field output factor. This correction factor is involved for dose determination in small fields realized on specialized systems (such as Gamma Knife, TomoTherapy, CyberKnife, etc.) as well as for small fields defined on conventional linacs. The correction factor $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ provides a link to dose based on ratios of detector readings as defined in Equation 4 and differs from unity for the majority of the detectors.¹

The calculation of $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ for several detectors in small fields has been the subject of many investigations.^{50,55,65-87} For air-filled ionization chambers, $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ can theoretically be expressed as the ratio of the product of the field size-dependent water-to-air restricted stopping-power ratios, $[\bar{L}/\rho]_{air}^w$, and the overall chamber perturbation correction factor, p , for the clinical (f_{clin}) and the machine-specific reference (f_{msr}) field sizes, respectively, as shown in Equation 5.⁸⁸

$$k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}} = \frac{\left[\left(\frac{\bar{L}}{\rho} \right)_{air}^w \cdot P_{fl} \cdot P_{gr} \cdot P_{stem} \cdot P_{cel} \cdot P_{wall} \right]_{f_{clin}}}{\left[\left(\frac{\bar{L}}{\rho} \right)_{air}^w \cdot P_{fl} \cdot P_{gr} \cdot P_{stem} \cdot P_{cel} \cdot P_{wall} \right]_{f_{msr}}} \quad (5)$$

$P_{gr} \equiv P_{\rho} \cdot P_{vol}$ accounts for the perturbation due to mass density and volume effects.⁸⁸ Although an expression as in Equation (5) can provide insight in the detector-associated physical reasons for the sometimes substantial correction factors in small-field dosimetry, most calculated datasets are determined by using direct MC calculations of the double ratio expressed in Equation (4). The mean values of $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ averaged over full-width at half-maximum (FWHM) and energy for various microdetectors have been provided for Siemens, Elekta, and Varian machines by several investigators.^{72,76,89}

Alternatively, the $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ can be derived by experimentally comparing the detectors' response to that of a reference detector with known response in varying field sizes.^{77,90,91} For the same nominal energy, the $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ factor mostly depends on the field size, expressed by the FWHM of the dose profile at the depth of measurement, and the type of detector. It is less dependent on the linac model, the radial FWHM of the primary electron source or the energy of the primary electron source, the distance between the exit window, and the target. Therefore, it has been suggested that a mean value of $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ could be used with an acceptable uncertainty.^{72,76,89} Thus, the numerical values of $k_{Q_{clin},Q_{msr}}^{f_{clin},f_{msr}}$ cannot be applied to other types of detectors and machines of differing nominal energies. Of all these parameters the field size at the point of measurements, f_{clin} , is critical and needs to be determined experimentally. In

TRS-483, all field output correction factors are specified as a function of the field size expressed by the parameter S_{clin} which is determined by explicitly measuring the FWHM of the profiles both in-plane (x) and cross-plane (y) direction at the depth of output determination as given in Equation 6.

$$S_{clin} = \sqrt{FWHM(x) FWHM(y)} \quad (6)$$

Please note that this definition is applicable only in small fields whereas in broad beams, the equivalent square field is chosen such that it provides equal photon scatter contribution. Additionally, this definition is not fully verified and need additional investigations. For specialized machines like Gamma Knife, CyberKnife, and TomoTherapy, users should refer to specific data collection using other types of detectors for which the detector-specific correction factor is known, such as reported by TRS-483.¹ A number of other investigators have provided data available in the published literature for comparison and reference^{69,71,73,76,89,92–97} that continues to be an active area of investigation.

3 | SMALL FIELDS: DEFINITION OF FIELD SIZE

Small radiation fields have been defined as those that satisfy one or more of the following conditions: (a) LCPE lost on the beam axis, (b) collimating devices partially occlude the primary photon source in the beam axis, or (c) the detector size is large or similar to the beam dimensions.^{1,30,31,51} The definition of a small radiation field is also subject to the detector characteristics with respect to the lateral range of the charged particles produced in the photon beam. Due to electronic disequilibrium, there is a reduction in dose at the center of the beam with decreasing field sizes^{28,29,98} as shown in Figure 2.⁹⁹ Source occlusion or blocking of the direct beam source leads to a reduction of the primary fluence resulting in a drop in beam output and the overlap of penumbras formed from opposing collimating jaws (Figure 3). The dimension of the primary photon beam source (also known as the focal spot) is on the order of millimeters and varies with machine type and accelerating potential. Figure 4 shows a wide range of source sizes determined for older machine types and for new machines.¹⁰⁰ On the modern machines such as the Varian TrueBeam, the focal spot has been shown to be relatively constant (<1.0 mm) for most energies and to have submillimeter (0.7–0.9 mm) dimensions in the x and y directions, as shown in Figure 4.¹⁰¹

The size of the focal spot of a linear accelerator can be determined by several methods and the interested reader is referred to the information provided in the references.^{100–110} However, as subsequently noted, it is

not critical in clinical practice and it is not necessary to measure the focal spot for clinical applications.

The signal measured by the detector in a small field depends predominantly on the size of the detector's active volume in relation to the field's dimensions, its construction, the densities of its active volume and other components.¹¹¹ As a result, perturbation of charged particle fluence can lead to an incorrect interpretation of the measured data.¹¹¹ Li et al.²¹ demonstrated that complete LCPE could be achieved if the photon beam radius is greater than the range of secondary electrons to achieve lateral electronic equilibrium (r_{LCPE}), and provided a relationship between the beam quality index in terms of TPR_{10}^{20} and r_{LCPE} in water. If $\%dd(10)_x$ is used as the quality index as defined in TG-51,⁴ the approach by Kalach and Rogers¹¹² can be used to convert TPR_{10}^{20} to $\%dd(10)_x$. Table 2 shows values of r_{LCPE} for the typical photon energies used in radiation therapy based on equation provided by Li et al.²¹ The values in Table 2 may differ from more advanced calculations as provided by Papaconstadopoulos¹¹³ and adopted in TRS-483¹ for r_{LCPE} as given in equation 7 in terms of TPR and equation 8 in terms of $\%dd(10)_x$. It is noted that the value of r_{LCPE} will be slightly different (~1 mm) between these two equations.

$$r_{LCPE}(cm) = 8.369 \times TPR_{20,10} - 4.382 \quad (7)$$

$$r_{LCPE}(cm) = 77.97 \times 10^{-3} \times \%dd(10)_x - 4.112 \quad (8)$$

It should also be noted that Equations 7 and 8 are meant to provide an approximate value of r_{LCPE} for a flattened beam, for the purposes of estimating when a beam is small and should be used with caution when FFF beams are considered. For FFF beams, the $\%dd$ and TPR values are substantially lower in comparison with the filtered beams of the same nominal MV value, while their r_{LCPE} is similar to that of a flattened beam at the same nominal accelerating potential.

Equations 7 and 8 can be further simplified when TCPE exists, for example, where dose in water (D_w) and collision kerma (K_{coll}) are identical. Given these approximations for clinical radiation therapy applications, the range of r_{LCPE} values can then be represented in terms of d_{max} of the beam for a clinical physicist:

$$r_{LCPE}(cm) \approx 0.67d_{max} \pm 0.2 \quad (9)$$

where d_{max} is the depth of maximum dose in cm for the 10 cm × 10 cm reference field. The range of the lateral travel of secondary electrons and its relation to LCPE becomes more prominent in inhomogeneous media, especially in a low-density medium such as lung.^{111,114–116}

In summary, a photon field should be considered a small field when the distance from the central axis to the field edge is smaller than the lateral range for

charged particle equilibrium. In practice, this distance should be measured with respect to the outside dimension of the detector. As this task group provides recommendations on dosimetry in 6 MV photon beams, fields $\leq 3 \text{ cm} \times 3 \text{ cm}$ are typically considered as small fields. A field should be considered a small field if any of the following is true:

1. Field size (diameter) $< 2 \cdot r_{LCPE} + d$, where d is the maximum detector outside dimension¹
2. Detector's view of the primary photon source is obstructed by the primary collimator
3. The detector composition, size, or design is unsuitable for optimal readings in the measurement conditions (Section V)

4 | DETECTORS

The value of the r_{LCPE} greatly restricts the physical dimensions of the detectors that can be used for the experimental determination of small field dosimetric parameters such as field output factors (Ω s), beam profiles, PDD, TMR, or TPR.^{34,37,117,118} When used for measurements in small fields, a detectors' reading must be adjusted for perturbation of particle fluence caused by the physical size, densities of the active volume, surrounding components, and non-water equivalence of the dosimeter.¹¹⁹⁻¹²¹

The Proceedings of the AAPM Summer School 2009¹²² include a comprehensive discussion on

various radiation detectors; however, these are not specific to small-field measurements. Specific guidance on the choice of detectors for use in small-field dosimetry can be found in IPEM Report 103³⁰, AAPM Report TG-120²⁰ and ICRU-91,⁹ which include a list of detectors and their characteristics in small field and IMRT dosimetry. Also a comprehensive summary on detectors applicable for small-field reference and relative dosimetry is included in the TRS-483 report.¹

The characteristics of a detector including the value of $k_{Q_{clin}, Q_{ref}}^{f_{clin}, f_{ref}}$ and its constancy near unity is desirable as a function of clinical conditions but this is hard to achieve. There are only a few detectors (plastic scintillators, microDiamond, some diodes, and Gafchromic films) that are suitable for small-field dosimetry and have a near-unity $k_{Q_{clin}, Q_{ref}}^{f_{clin}, f_{ref}}$ factor. The characteristics of these detectors have been summarized in the literature.¹²³⁻¹²⁵ This section presents a brief summary of the most relevant detectors and gel dosimeters for current and future applications.

Ionization chambers are the most commonly used detectors in radiation dosimetry due to their favorable characteristics (high precision, linearity, and stability). However, small-volume ionization chambers experience low signal-to-noise ratios and in some cases high polarity effects. Solid-state dosimeters such as diodes and metal-oxide-semiconductor field-effect transistor (MOSFET) have smaller sensitive volumes. However, these dosimeters are not water-equivalent and exhibit a differential energy dependence with respect to

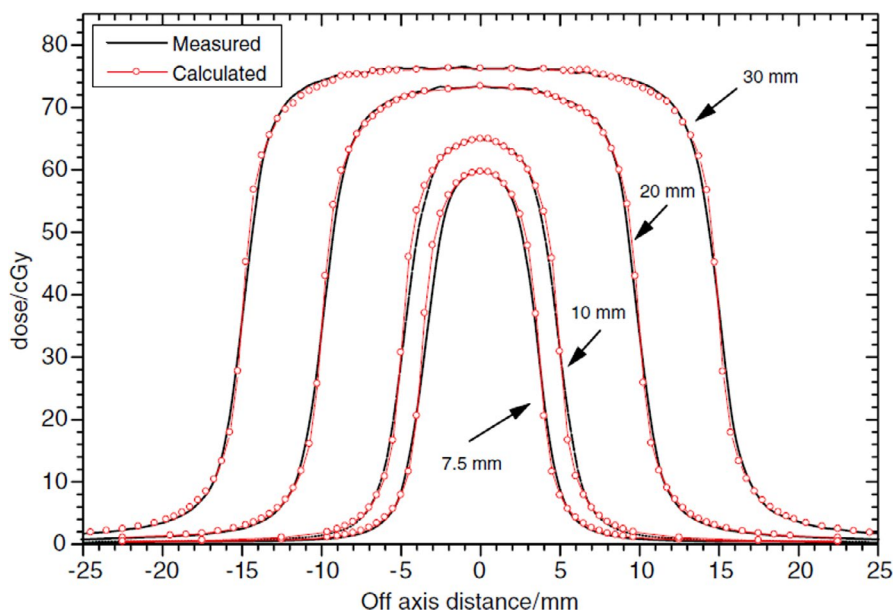


FIGURE 2 Comparison between Monte Carlo simulated and experimentally measured dose profiles for 6 MV at 1.5 cm depth in water. Note the reduction in dose at the central axis as the field size is reduced. The reference beam (10 cm $\times$ 10 cm) was calibrated according to the AAPM TG-51⁴ dosimetry protocol to give 1 cGy/MU at 100 source-to-surface distance (SSD) at a depth of 1.5 cm. The Monte Carlo simulation was also normalized to provide 1 cGy/MU for the same reference condition. For a large field, the magnitude is 100% (not shown in the figure) whereas for a small field (shown in the figure), the output is low (55%) Reproduced from Ding et al⁹⁹ [Color figure can be viewed at wileyonlinelibrary.com]

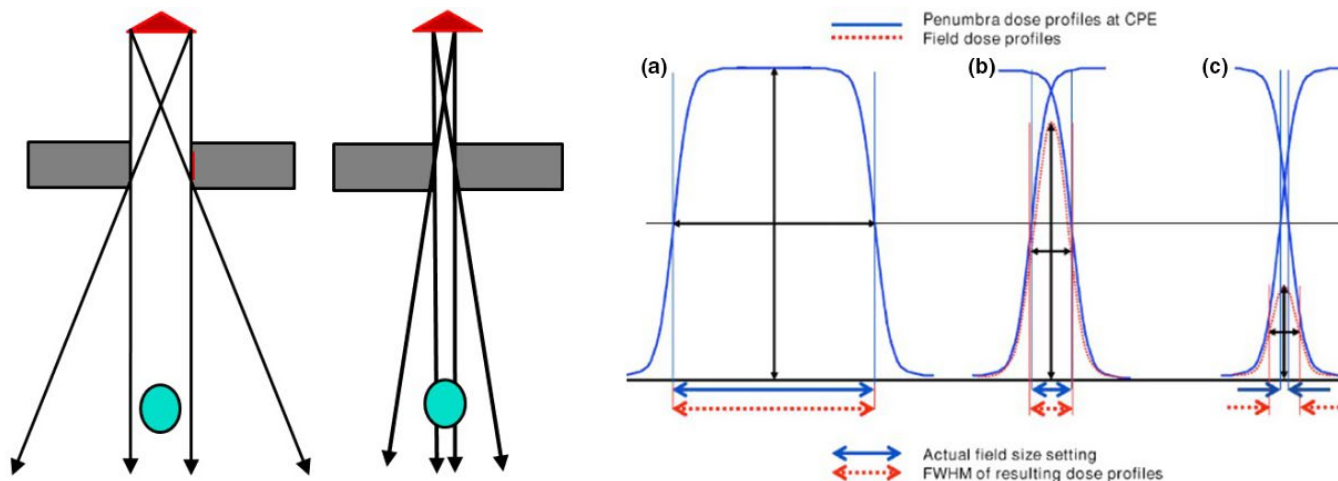


FIGURE 3 Source size and detector size with respect to collimator setting. Left: When the field size is reduced, it obstructs the source and limits the photon fluence. Right: As the field size is reduced, the penumbrae from opposing jaws overlap and the dose drops at the center of the field. As a result, the FWHM of the dose profile is no longer equal to the collimator setting. (Adapted from Das et al.⁵¹) [Color figure can be viewed at wileyonlinelibrary.com]

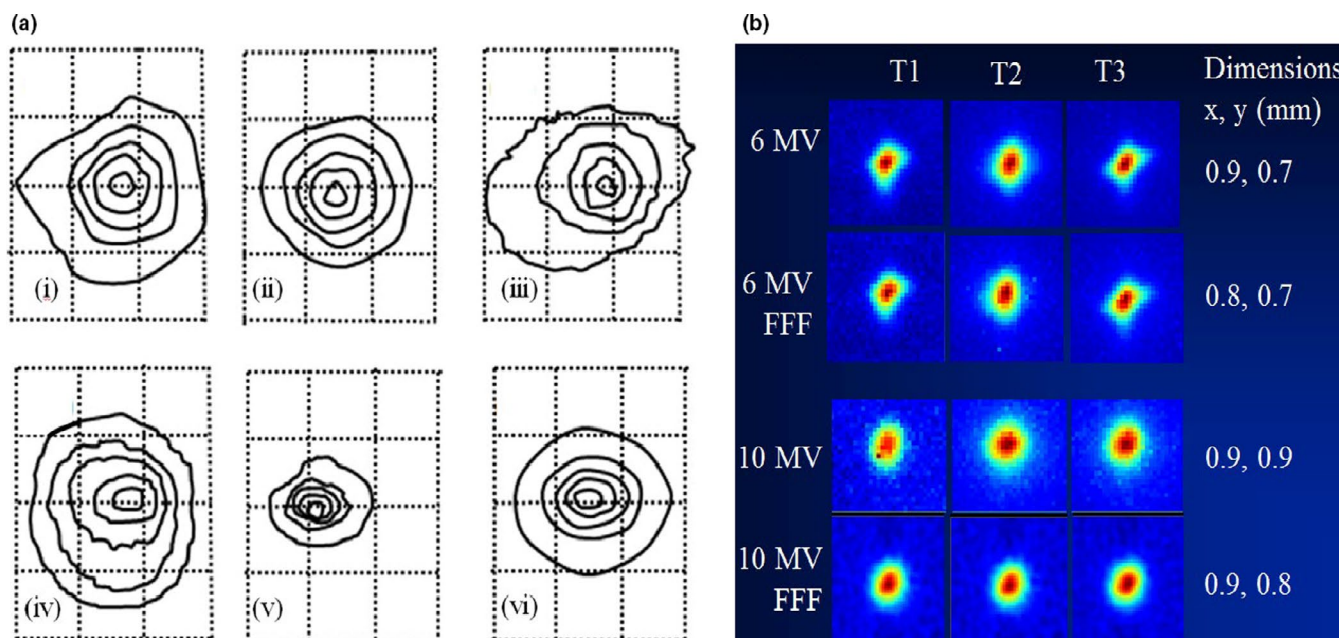


FIGURE 4 Iso-intensity and focal spots of Varian and Siemens accelerators. (a) Iso-intensity plots of the focal spots from Varian and Siemens accelerators. The intensity distribution shown corresponds to 90%, 70%, 50%, 30%, and 10% intensities. The 90% intensity corresponds to the innermost line and the 10% intensity corresponds to the outermost line with other intensity lines in between. (i) Varian 2100C, 6 MV, (time point 1); (ii) Varian 2100C, 6 MV, (time point 2 after 2 years); (iii) Varian 2100C, 18 MV, (time point 1); (iv) Varian 2100C, 18 MV, (time point 2 after 2 years); (v) Siemens KD-2, 6 MV; and (vi) Siemens KD-2, 23 MV. The square grid size is $1 \times 1 \text{ mm}^2$. Additional focal spot images can be viewed in the reference by Jaffray et al.¹⁰⁰ (b) Focal spot from three True Beams (T1, T2, T3) for 6 and 10 MV with and without FFF. The dimensions in x and y directions are noted. Modern machines such as the Varian TrueBeam have submillimeter focal spot sizes in both dimensions.¹⁰¹ Image reproduced from Sawkey et al.¹⁰¹ [Color figure can be viewed at wileyonlinelibrary.com]

water. Thus, their response with field size and the possible effects of beam hardening at the measurement depth must be considered. References [119, 126–129] showed that for small fields the dose absorbed by the detector's sensitive volume depends upon its density. High-density detectors over-respond while low-density detectors under-respond.¹³⁰ Correction factors

are often required to convert the measured detector signal to an absorbed dose. As no single detector ensures accurate dose ($\pm 2\%$) determination for field sizes ranging from $0.4 \text{ cm} \times 0.4 \text{ cm}$ to $3 \text{ cm} \times 3 \text{ cm}^2$ the current consensus recommends making a careful comparison using at least two types of acceptable detectors.^{43,126,127}

Beam	TPR _{20,10}	%dd(10 cm) _x	r_{LCPE} (cm)		
			TPR _{20,10} (Equation 6)	%dd(10 cm) _x (Equation 7)	d_{max} (Equation 9)
Co ⁶⁰	0.579	58.7	0.5	0.5	0.5
6 MV	0.670	66.2	1.2	1.1	1.1
10 MV	0.732	73.5	1.7	1.6	1.7
15 MV	0.765	77.9	2.0	2.0	2.0
24 MV	0.805	83.0	2.4	2.4	2.5

TABLE 2 Values of r_{LCPE} for various beams calculated with the Monte Carlo method, where r_{LCPE} is the minimum beam radius to achieve complete lateral electron equilibrium in water²¹

4.1 | Real-time detectors

4.1.1 | Ionization chambers

Air-filled ionization chambers are most commonly used for dosimetric measurements because of their high sensitivity, long-term stability, reproducibility, and robustness. These characteristics enable ionization chambers to have calibration coefficients that are traceable to national standards. Two types of ionization chambers should be distinguished: cylindrical (known as “thimble” chambers) and plane-parallel. The sensitivity of air-filled ionization chambers varies in direct proportion to its sensitive volume. Very small-volume chambers produce very small ionization currents. Despite this limitation, small-volume ionization chambers have been used successfully for dosimetric measurements with substantial correction factors^{38,40,50,117,131} for fields down to 1 cm × 1 cm.

4.1.1.1 | Signal-to-noise ratio and charge leakage

Small-volume ionization chambers are used to improve the spatial resolution of the measurements. The central electrode (0.55–0.16 mm in diameter) has been sometimes constructed from an electrode of high atomic number (Z) material (e.g., steel) in order to increase the signal from their small cavity. MC simulations in a broad field and at depths down to 20 cm in water have shown that the high-Z material in the chamber creates perturbations up to 3% relative to detectors having a central electrode made of a low-Z material.¹³² In the case of narrow fields, such perturbations have been shown to be significant. For this reason, in small-field dosimetry the use of chambers with high-Z material should be avoided,^{20,131} whereas ionization chambers with an aluminum central electrode do not exhibit large perturbation effects.

Due to the low sensitivity of small volume ionization chambers, a measurement that does not consider leakage can result in an error in absorbed dose of up to 16%, depending on the irradiation technique.¹³³ The magnitude of the leakage for the chamber-electrometer combination can be measured, assuming that the leakage current is constant and does not change when the beam is on. However, the background signal while the

beam is on could be an issue when using a microchamber as shown by Sarkar et al.¹³⁴ In general, for small microchambers background readings that would be collected over the same time as the irradiation time should be subtracted. The extracameral signal increases with the length of irradiated cable in most chambers.^{5,131} Leakage current introduced by the cable, connectors, and electrometer can have a larger effect on small-volume ionization chambers due to their smaller signal-to-noise ratios than for the larger volume Farmer type ionization chambers.

Another approach to overcome the challenge of low signal-to-noise ratio and to increase the sensitivity of small-volume chambers is to replace the air cavity with a dielectric fluid. Liquid-filled ionization chambers (LICs) have lower perturbation effects due to the near water equivalence of the dielectric liquids.^{135,136} These enable the acquisition of measured datasets of higher spatial resolution (<0.1 mm). However, corrections for ion recombination effects are substantial, and the standard method of correcting ionization recombination does not apply to liquid-filled cavities because the collected charge does not increase linearly with voltage and only saturates at voltages too high that are suitable for the insulators used in the detector.^{137–140} Chung et al¹⁴¹ have proposed an accurate method to correct for the general recombination in this detector. Nevertheless, liquid-filled ionization chambers,^{72,137,142,143} despite being promising for small-field dosimetry applications, are no longer commercially available.

4.1.1.2 | Volume averaging

Depending on the field size and dimensions of the ionization chamber, the detector's signal may need to be corrected for volume averaging. The correction value can be calculated using the beam profiles at the measurement point in the radiation field. It is determined by taking the ratio of absorbed dose to water at the point of measurement without the detector to the absorbed dose to water in the phantom averaged over the detector's sensitive volume without the detector. Profiles acquired for treatment planning commissioning should be corrected for volume averaging, or, preferably, use a small enough detector with minimum perturbation such that volume averaging is unnecessary.

4.1.2 | Solid-state detectors

The high sensitivity of diodes permits their construction with very small dimensions, a favorable characteristic for use in small-field dosimetry. The active volume of a diode is determined by the depletion region, which usually ranges between 20 and 80 μm . Diodes may need additional corrections for the following characteristics: dose rate dependence, variation of response with accumulated dose (up to 10%), temperature dependence ($\sim 0.3\%/^{\circ}\text{C}$), contact material, energy, and angular dependence.¹⁴⁴

The restricted mass collision stopping-power ratio of water to silicon, $[\bar{L}/\rho]_{\text{Si}}^{\text{w}}$ in small fields can be assumed to be close to the values at the reference field (10 cm $\times$ 10 cm), but this is not the case for the spectrum-averaged mass energy-absorption coefficient ratio of water to silicon $[\bar{\mu}_{\text{en}}/\rho]_{\text{Si}}^{\text{w}}$.^{41,145} The atomic number and density of silicon are different from that of water thus greatly affecting the fractional response due to photoelectric interactions. This effect is prominent in large fields as the low-energy photon scatter is substantially contributing to the spectrum, but this is relevant for small-field dosimetry field output factors as detector readings are normalized to those in the reference (large) field. In small fields, an additional opposing effect is due to the presence of silicon, which introduces an over-response in small fields due to a fluence perturbation that is not negligible.^{146,147} A possible solution is to use the intermediate field method, sometimes referred to as the daisy chain technique^{148–150} (cross calibration of diode against the chamber in an intermediate field to connect measurements in large fields to small fields) to measure field output factors normalized to a 10 cm $\times$ 10 cm field. Although this technique accounts for the energy dependence of large fields, it does not consider the electron fluence perturbation due to the high density of silicon in small fields. A multi-institutional investigation from Japan for small-field dosimetry with high definition MLC demonstrated large variations in the dosimetric parameters, for 5 mm square fields and it was attributed to measurement errors due to selection of detectors.¹⁵¹ It should be noted that even with the use of the intermediate field method, each of the detectors involved in the process must be corrected for the appropriate field output correction factors.

Mobit et al¹⁴⁶ showed that Spencer-Attix cavity theory underestimate dose in high density cavity and provided Monte Carlo (MC) modeling of the energy response of the diodes in the detector cavity which included an energy-dependent factor to compensate for the lack of local charged particle equilibrium that account for depth and field size dependence of the diode. Scott et al.¹¹⁹ demonstrated with the MC method that diodes could be used to determine the dosimetric parameters of small fields down to 1 cm $\times$ 1 cm. Simulations and

measurements showed up to 4.5% differences in fields smaller than 1 cm $\times$ 1 cm, and they were attributed to the source occlusion and the difficulty of accurately modeling the focal spot.

4.1.2.1 | Diode types

Three common types of silicon diode designs are commercially available: an unshielded diode (such as the PTW-60017) generally recommended for electron beams; a shielded diode (such as the PTW-60016) containing a tungsten-epoxy filter to decrease the fluence of low-energy photons, and an unshielded stereotactic photon diode (such as the IBA SFD) specifically designed for measurements in stereotactic beams. Details on the characteristics of recommended diode detectors are provided in TRS-483¹ and IPEM report 103.³⁰ Numerous publications recommend not using the shielded diode detectors for small-field measurements because the shielding introduces additional scatter and increased directional dependence.^{122,146,147,152} In contrast, several other publications reported $<1\%$ correction factors for a shielded diode in small fields.^{153–155} Due to the conflicting published information regarding the use of a shielded diode in small-field measurements, this report recommends the use of either a stereotactic or an unshielded diode since these latter diodes yield similar results in small-field measurements (see data from TRS-483¹). In any situation, correction factors are required when the fields are small for most diodes.

Recently, a new unshielded silicon diode detector (PTW-60023) known as a microSilicon detector with very small dimension (0.032 mm³ with 1.5 mm diameter) with an epoxy density of 1.15 g/cm³ was introduced and was found to have very suitable characteristics for small-field dosimetry for both linear accelerators and CyberKnife.^{156–159} Even the new shielded diode (PTW-60022) showed a very good characteristics and superior results in very small photon fields.¹⁶⁰

4.1.3 | Diamond detectors

Diamond detectors have been well characterized.^{45,161–167} One of the main advantages with the natural diamond detector for measurements in small fields is that it is nearly tissue equivalent due to the low atomic number, $Z = 6$, which is close to that of water, $Z = 7.4$. It offers relatively high spatial resolution with high sensitivity. Considerations hindering their widespread use are: (a) the mass density of diamond, 3.5 g/cm³, is higher than that of water; (b) detector dimensions (including the materials around the active layer) can be relatively large for use in very small fields such as 1 cm $\times$ 1 cm; (c) natural diamond detectors are very expensive; (d) diamond detectors have a dose rate-dependent response^{161,164–166}; (5) there is large variability in the

dimensions of the active layer among detectors of the same type; and (6) as a result, correction factors are specimen dependent.^{163,168} Thus, a single, universal, correction cannot be recommended for all diamond detectors of the same type. The issues mentioned above appear to be overcome with the introduction of synthetic chemical vapor-deposited (CVD) diamonds.¹⁶⁹ However, as in the case of solid-state detectors, in small-field sizes (less than 1 cm × 1 cm) a CVD would also introduce photon fluence perturbations due to the difference of its mass density from that of water and would require appropriate corrections. Detectors with the early design using a natural diamond should be used only with proper MC simulations or published and inter-compared $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ factors.

With advances in crystal design, a single-crystal diamond detector (known as the microdiamond detector) is commercially available. Due to its tissue equivalence and very small size, this detector has been shown to measure dose with only a small perturbation for very small-field sizes. Several groups have shown that $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ can be treated as unity in small fields.^{73,76,96,170,171} However, data presented in the TRS-483¹ report show small correction factors below 1.5 cm², whereas for the natural diamond detector the correction factor is close to unity (±1%) for very small fields. De Coste et al.⁷⁸ systematically characterized the PTW microDiamond field output factors. To better understand its response in small fields and to investigate its suitability for reference dosimetry, 10 microDiamonds were calibrated under ⁶⁰Co irradiation. The field output factors measurements were performed in 6 MV photon beams from different manufacturer (CyberKnife, Varian, Elekta) linacs. They demonstrated that the detector response for field sizes larger than approximately 1 cm × 1 cm did not require corrections, since $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ is within 1% of unity. Das and Francescon¹⁷² indicated that for very small fields (≤0.75 cm) $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ may differ from the values published by Azangwe et al.⁷⁷ which was adopted in TRS-483.¹ A recent publication by Casar et al.¹⁷³ measured field output factors for a microDiamond and six other diode detectors and reported a systematic over-response of the microDiamond detector for fields ≤1 cm × 1 cm which differed from the data provided in TRS-483.¹ It was suggested that correction factors for this detector depend upon the type of linac, beam energy, and collimation used. The behavior of correction factors for the microDiamond in very small-field sizes (≤1 cm × 1 cm) was shown to depend on the combination of volume averaging and perturbation effects in the detector material. These two relevant effects compensate each other very well in the microDiamonds dosimeter, leading to relatively small overall output correction factors. As a result, correction factors lower than 1.5% were evaluated for beam sizes down to 5 mm in all linacs.⁷⁸ Because similar curves of $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ were observed for the three different linacs used in this study, a single

empirical function was calculated that provides output correction factors within 0.5% from the MC values derived for all three linacs, down to fields as small as 4.6 mm in size. This maximum deviation was shown to be comparable to the intrinsic repeatability for the very small-field sizes obtained by linacs using jaws and/or MLCs. Because of the conflicting data, users of the microDiamond dosimeter are encouraged to evaluate the detector performance and to compare it to the newly published data^{78,79,86,172,174–177} that may differ from that of TRS-483.¹ Also, as recommended in TRS-483, at least one other suitable small-field detector should be used to compare with the microDiamond measurements.

4.1.4 | Plastic Scintillators

Plastic scintillators have a number of favorable dosimetric characteristics for small-field dosimetry, such as their relative tissue equivalence, angular independence, stable photon energy response, small size, density similar to water, and high sensitivity.^{90,178–184} Morin et al.¹⁸³ compared small-field measurements with plastic scintillators and various diodes for a CyberKnife. They reported good agreement between the scintillator measurements and MC calculations. Due to their unique characteristics, this dosimeter can be used for small-field dosimetry without any correction factors.^{1,185} However, the development and clinical implementation of plastic scintillators are still a work-in-progress. There is only one such detector (PSD-W1) available from a single manufacturer (Standard Imaging, Middleton, WI). The use of the detector is not as straightforward as with other detector systems. Its use requires a certain irradiation geometry in a solid phantom, use of a specific type of readout system, and corrections to account for the additional signal (light) that is generated by the system itself (Cherenkov radiation in the light fiber).¹⁸⁶ Even though, $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ is 1.0 for most field sizes as reported in TRS-483, the literature is not uniform about W1 being a problem-free detector. This is due to Cherenkov radiation in the fiber in the commercial W1 as the calibration procedure involves a particular irradiation geometry of the fiber that does not necessarily account for a small-field application. The specific angular distribution of the Cherenkov radiation during irradiation in small fields, with the detector oriented vertically, led to systematic effects on the data in the smallest fields.^{84,184} This behavior prevents the use of the W1 for scanning. To overcome some of the problem associated with W1, Standard imaging introduced PSD-W2 in 2019 with the added capability for using it in a scanning system. The PSD-W2 has a better system to counteract and mitigate the Cherenkov radiation. The characteristics of the W2 are identical to that of W1 as has been reported by Galavis et al.¹⁸⁷ In addition to the W1 and W2 plastic scintillators, inorganic scintillator detectors of about 200 μm diameter are being studied that could be

used for ultra-high-resolution dosimetry as shown by Debnath et al.¹⁸⁸

4.2 | Passive Detectors

4.2.1 | Two-dimensional passive detectors

Radiochromic films^{123,189–192} provide two-dimensional (2D) dosimetric information, not available immediately following irradiation; and for this reason they are referred to as 2D passive detectors. Because of their superior spatial resolution compared to other detectors, they can provide unsurpassed results for beam profiles and FWHM measurements. Radiochromic film can be used as well for field output factor measurements. Radiochromic film has an effective atomic number and density similar to water. It exhibits an approximately linear dose response and does not require chemical processing.¹⁸⁹ Detailed information on radiochromic film is available in the book by Das.¹²³ The optical density changes with time post-irradiation and its spatial nonuniformity can produce a position-dependent variation in its radiation response.^{47,193} The readout system could be relatively expensive, although inexpensive flat-bed color scanners can be used, provided that stringent protocols are followed in the readout procedure.¹⁸⁹ Numerous publications^{193–196} have reported dose accuracy within ± 2 –3% in small fields. However, the dosimetric characteristics must be assessed when the manufacturer makes a significant change in the emulsion formula and it becomes a new product. This re-characterization may be time-consuming for clinical use. For small fields, there are many controversial statements made for its handling and usage as reported in various publications.^{197–201} In view of this controversy, it is advisable to strictly adopt all the precautions needed for radiochromic film dosimetry.^{123,192}

4.2.2 | Luminescent dosimetry

Thermoluminescent detectors (TLDs), radiophotoluminescent glass (RPG), and optically stimulated luminescent detectors (OSLDs) are passive detectors that can be used only for point dose measurements.²⁰² TLDs come in various sizes and shapes such as powder, chips, rods, and microcubes. Depending on the composition of the TLD, they have a wide dose range and are nearly water-equivalent.²⁰³ Their response is relatively energy independent for megavoltage photons and electrons, and they exhibit only a small angular dependence. However, unless individual detector sensitivities are determined, they have a relatively large uncertainty, about $\pm 3\%$, and measurements are labor-intensive.²⁰⁴ The uncertainty for relative measurements can be reduced to 1.0%

by carefully determining individual relative sensitivities by calibrating them in a uniform radiation field and maintaining good quality control on the readout method.^{205,206} Regardless, TLDs have been used for small-field measurements.^{39,193} Massillon et al.¹⁹⁶ showed that with proper care, TLDs can be used for small-field measurements. Published studies^{77,207} report that TLDs, especially 1 mm $\times$ 1 mm $\times$ 1 mm microcubes, can be used with about a 1% correction factor for field sizes down to 0.6 cm $\times$ 0.6 cm. Detectors such as RPG^{48,208–210} and OSLD^{211,212} have dosimetric characteristics similar to TLDs for megavoltage photon and electron radiation except that light is used instead of heat to eject the trapped electrons to recombine with holes to emit light. RPGs and OSLDs have a similar linear dose response and are easier to handle compared to TLDs. They can be re-read since the radiophoto luminescent centers remain constant for RPGs, and each readout procedure of the OSLD detectors decreases the signal by only about 0.05%. However, unlike TLDs, optical bleaching of OSLDs does not eliminate all of the radiation-induced effects. For accumulated doses greater than 20 Gy, there is an increase in the detector sensitivity and background signal. As with TLDs, the response is relatively independent of megavoltage photon and electron beam energies and exhibits only a small angular dependence.^{48,210,213,214} They can be readout 10 minutes after irradiation. Commercially available OSLD using $\text{Al}_2\text{O}_3\text{:C}$ can be placed in water for up to three hours without any degradation in its properties. However, it must be shielded from light, which can reduce the radiation-induced signal. These types of dosimeters as well as their specialized readout systems have recently become available and their dosimetric properties appear promising for use in small-field dosimetry.²¹¹ As with TLDs, individual relative sensitivities should be determined to reduce the measurement uncertainty to $\pm 2.0\%$. Since the mass collision stopping-power and the mass energy-absorption coefficient ratios of water to LiF or Al_2O_3 are relatively constant for energies between 0.1 MeV and 10 MeV, it is expected that perturbation effects would be negligible for small detectors. However, the higher density of these detectors, especially OSLDs, causes electron fluence perturbation in small fields. The AAPM TG-191²⁰² has compiled guidance and recommendations on the clinical use of these detectors including their potential use for small-field dosimetry. Correction factors for TLD microcubes and chips are also provided in the TRS-483 report.¹

4.2.3 | Polymer gel dosimetry

Gel could be potentially useful for small-field dosimetry because the following common sources of error can

be eliminated or reduced: errors arising from aligning small detectors to small fields at all depths, volume averaging, and directional dependency. Furthermore, complex corrections to account for changes in media are not required. However, use of this techniques has been largely confined to academic institutions to date. Two main categories of three-dimensional (3D) dosimetry systems are described below. Due to the limited number of clinical users, this class of detectors has not been described in recent international reports.^{1,2,9}

Polymer gel dosimeters consist of a firm gelatin gel doped with a monomer compound. They respond to radiation by a polymerization reaction which forms polymer microparticles in proportion to the delivered dose.²¹⁵ The polymer microparticles remain fixed in position within the gel matrix and the distribution of particles can be imaged by MRI, optical computed tomography (optical-CT), or x-ray CT to provide information on delivered dose.^{216,217} For small-field dosimetry, optical techniques have an advantage of high spatial resolution and better signal-to-noise ratio compared to MRI readouts.²¹⁸ The polymer gel dosimeters (MGS Research Inc., Madison, CT) with Z_{eff} of 7.4 and density of 1.02 g/cm³ provide an excellent medium for water-equivalent measurement. Polymer gels are used for a wide range of dosimetry measurements in radiation therapy including small-field dosimetry.^{90,219,220} With careful attention to readout and calibration procedures as outlined by Baldock et al,²¹⁷ isotropic 1 mm 3D dosimetry is feasible within $\pm 3\%$ uncertainty.

4.2.3.1 | Radiochromic gel or plastic dosimetry

Two types of radiochromic 3D dosimeters are in current use: radiochromic plastic (PRESAGE[®], Heuris Pharma, Skillman, NJ)^{221–223} and radiochromic gels (e.g., Fricke or Fricke-Xylenol-Orange, FBX gels).^{224,225} Both materials have good tissue equivalence, provide a linear optical response to radiation, and can be read out using optical-CT techniques. These materials lead to less scattered light compared with previous gel formulations which can enable a more accurate and faster readout.²¹⁶ A key concern of FBX gels is the significant fading (1–20 hours) due to the time dependent diffusion of ferrous or ferric ions in the gel. Thus, the dose distribution in FBX gels needs to be imaged shortly after capture of the signal. Fricke gels are considered relatively easy to manufacture and are used successfully. The PRESAGE[®] material does not suffer from diffusion, but it is not readily manufactured outside a specialized laboratory. PRESAGE[®] is insensitive to oxygen and exposure to the atmosphere and its hard plastic substrate negates the need for an external container (thus reducing edge artifacts) and facilitates molding and shaping of various sizes and shapes of dosimeter.

Several authors^{221,225} have demonstrated measurement of small-field output factors, PDD, and dose profiles with radiochromic gels and plastic. Good agreement is reported between these modalities at the ± 1 –3% level for small square fields above 5 mm in dimension. Agreement within 4% has been reported for ultra-small fields down to 1 mm.²²⁶

The current methods of 3D dosimetry are considered relative, and require a calibration procedure to convert the measured parameter to dose, although linear relationships have been widely reported for both polymer and radiochromic gels, and radiochromic plastic dosimeters. A good understanding of the performance and characteristics of the scanning system (either MRI or optical-CT) is also critical to achieve accurate results with all of the methods discussed above.²²⁷ For small field commissioning measurements, the absolute dose delivered to the dosimeter can be tailored to maximize signal-to-noise ratio and dynamic range of the scanner.²²⁸ For optical-CT readout, this dose should not exceed the range where scatter and cross artifacts have been observed.²¹⁶ Since gel dosimetry is still an active area of research and requires specialized readout equipment, further validation is needed to determine its appropriateness for small-field dosimetry.

5 | RELATIVE DOSE PARAMETERS AND MEASUREMENTS

5.1 | Percentage Depth Dose (PDD)

PDD data are used as input for the configuration of beam models in TPSs and for second check MU calculation software. To directly measure PDDs for small fields, the most suitable detectors are micro-ionization cylindrical and parallel-plate chambers, typically with a sensitive volume less than 0.02 cm³. TRS-483¹ has provided detailed discussion on the suitability of micro-ionization chambers in terms of their leakage, polarity effect, and P_{eff} for these measurements. Additional details are also available in various references for several beam energies.^{117,229,230} Shielded and unshielded diodes, including the smallest of the unshielded stereotactic photon diodes (SFDs), have also been suggested to be suitable for this measurement.^{129,160} However, the use of a shielded diode is discouraged because its tungsten-epoxy layer below the silicon chip, which corrects for the high-Z response of the diode to low-energy photons at large fields, introduces measurement artifacts in small fields.¹⁷⁹ Also, radiochromic film is suitable for PDD measurements as it does not pose problems due to volume averaging or non-water equivalence.^{126,231,232} However, the parallel irradiation of the Gafchromic[™] EBT films is discouraged by Fontanarosa et al.²³³ who observed unacceptable under dosage in

PDD at depths greater than 15 cm (about 5% at 20 cm). In general, the detectors that have smallest $k_{Q_{clin}, f_{max}}^{f_{clin}, f_{max}}$ factor (microdiamond, microsilicon, plastic scintillator, EBT film)^{78,85,97,123,156,158,186-188,192,234-236} are preferred to avoid the need to correct the PDD. If the institution is unable to justify acquisition of at least one of these systems, the method described by Francescon et al²³⁷ and as described in Figure 5 could be adopted.

The methodology for the experimental determination of PDDs in broad fields is discussed extensively in TG-106.⁵ The PDD has traditionally been calculated as a ratio of the ionization readings, and this method is generally valid for large (>3 cm × 3 cm) fields, where there is no need to correct for varying perturbation effects at varying depths. However, using the ratio of the readings to determine the PDD for small fields might not be valid if perturbation corrections indeed vary with depth and field size and requires accounting for the variation in the measurements. Therefore, with the choice of an appropriate detector, the detector-specific correction factor, $k_{Q_{clin}, f_{max}}^{f_{clin}, f_{max}}$, where the field size at the position of the measurement increases is required. Francescon et al²³⁷ showed that the largest corrections are in the build-up region where microchambers under-respond by up to 10% and diodes over-respond by up to 3%. Beyond the build-up region, stereotactic diodes and microdiamond detectors reproduce the PDD in water to

within 2% up to a depth of 30 cm.²³⁷ Microchambers over-respond for the PDD at deeper depths. This effect is greater if the stem axis is perpendicular to the beam axis because the chamber in the plane perpendicular to the beam axis occupies a greater area. Microchambers can be used for PDD measurements for field sizes greater than 10 mm without applying a correction factor.^{1,85,86,238} If a microchamber is used, PDD measurements should be repeated with both positive and negative polarities and the results averaged. Also, the chamber leakage should be accounted for at various depths.^{5,239} Figure 5 provides PDD data from two acceptable detectors for CyberKnife small-field dosimetry. The data in the inset demonstrate that the PSD-W1 scintillator detector requires minimal corrections.

Seven major challenges need to be addressed for accurate PDD measurements of small beams: (a) choice of an appropriate detector, (b) alignment and orientation of the detector with central axis, (c) alignment of the detector axis to the beam's central axis at all depths, (d) positional accuracy defining the field, (e) placement of the reference detector within a small field, (f) monitoring machine output fluctuations, and (g) the physical size of a detector since it may not correspond to the size and position of its sensitive volume. Therefore, a beam profile must be measured to find

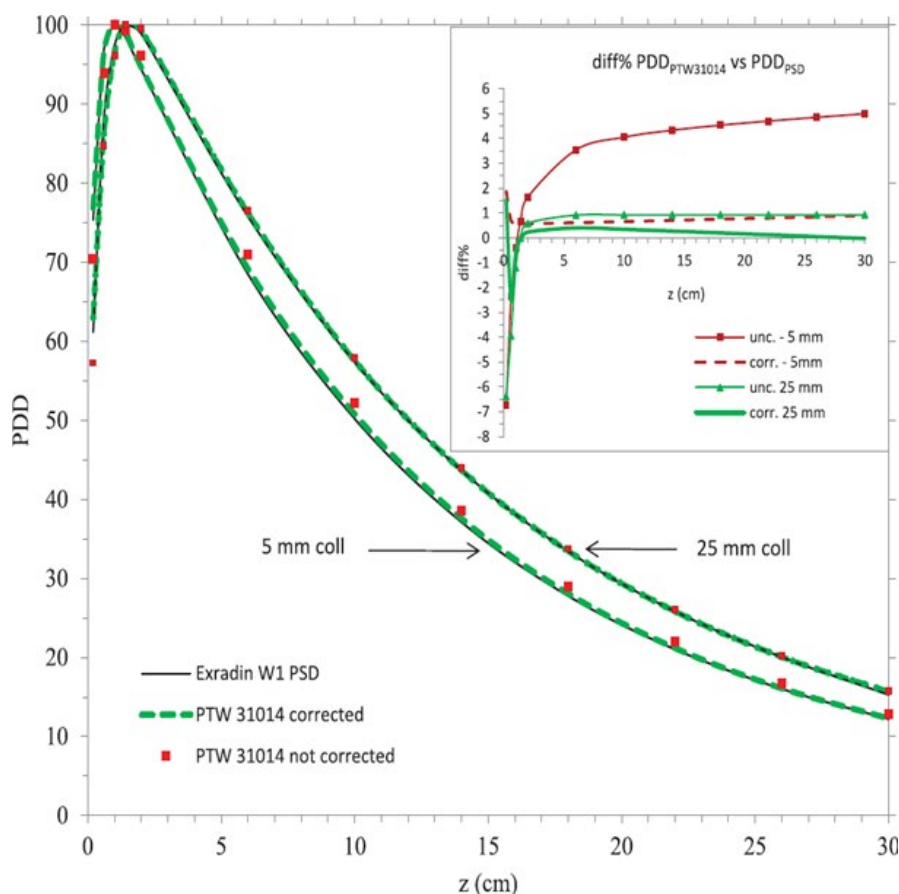


FIGURE 5 PDD of CyberKnife for 2 cones with two detectors. Inset is difference plot in dose compared with PSD detector that does not require correction. Adapted from Francescon et al²³⁷ [Color figure can be viewed at wileyonlinelibrary.com]

the position where the detector signal is maximum as described by Dietrich and Sherouse.¹⁴⁸ The effective detector size and central axis should be determined by scanning before measuring PDD or TMR. Also, the size of detector should be relatively small: its outer diameter should be less than half the field dimension to provide accurate data, as described in TG-120.¹¹

The choice of detector (item 1) for specific clinical needs has been discussed in the AAPM TG-106 report.⁵ The smallest unshielded stereotactic diode is recommended for these PDD measurements.^{129,179} To address the issue of detector alignment for small-field dosimetry (item 2), detector centering can be verified in a scanning water phantom by checking the symmetry of both orthogonal profiles at two different depths. Small-volume detectors (PinPoint, diamond, A-14 and stereotactic diode SFD) should be centered in a 1 cm × 1 cm field as shown by Dieterich and Sherouse.¹⁴⁸ An offset of 0.5 mm in the detector alignment can be easily detected in the depth profiles, either visually or by the symmetry of the profiles. Any correction to detector position with respect to central axis needs to be verified with additional orthogonal profiles. In general, the detector position can be aligned with the central axis to within a fraction of a millimeter. This method has been successfully employed in the determination of the zero-field TMR.^{240,241} Another method has been developed as well for detector alignment but may be more challenging to implement.²⁴² Finally, the orientation of the detector for scanning and PDD measurements can be important. TG-106⁵ and the TRS 483¹ reports recommend alignment of the detectors for these measurements. For ionization chambers the effective point of measurements will depend on the orientation of the chamber. Because of the small dimensions of the recommended ionization chambers for small-field measurements, the effective point of measurement shift will only have a minimal effect except in the build-up region with its high dose gradient.

The detector must be located along the beam's central axis for all depths so that the PDD is correctly measured. The effects of detector misalignment from the central axis (item 3) can be estimated by systematically shifting the center of the depth profiles in 0.2 mm steps in one direction for a 1 cm × 1 cm field and examining the change in symmetry of the profiles and the relative dose values of the profiles after the shift. The dose profile symmetry indicates whether or not the detector is aligned with the central axis. For example, the change in the relative dose is ≤0.5% for the 1 cm × 1 cm field within 1 mm of misalignment of the detector axis with central axis.²⁴⁰ To reduce errors from possible misalignment of the detector with the central axis, one can use the serpentine method to reconstruct a PDD curve from the maximum signal of narrow beam profiles measured at several depths as is performed in TomoTherapy.²⁴³

The effects of the accuracy of jaw settings (item 4) can be estimated by repeating the dosimetry measurements and calculations for a range of field sizes (e.g., increments of 2 mm; range 1 cm × 1 cm to 4 cm × 4 cm fields). For example, with a field size change of negative 2 mm, the zero-field PDD is found to increase by about 0.5%–2% over the range of depths 5–30 cm. However, a field size change of +2 mm results in a lower zero-field PDD by about 0.5%–1.0%.^{240,241} If measurements are made with a scanning system, the collimator settings can be verified by measuring dose profiles along the two major axes. For other systems, radiographic or radiochromic film can be used to verify the collimator settings and FWHM.

Placing a reference detector in a small field is a challenging task and should not be attempted during beam scanning (item 5). Transmission chambers are commonly available from most vendors (Stealth chamber: IBA; T-Ref: PTW; SNC-600C: Sun Nuclear)^{244,245} that can be used without any perturbations in small field scanning. If such a device is not available, the physicist should scan either without a reference chamber or with it placed at the bottom of the water tank with the gain increased to achieve a suitable signal. To monitor the machine output (item 6), measurements can be made with the detector under reference conditions at various time intervals to verify that the dosimeter response is reproducible over time or with a transmission chamber in the head under the collimator.

Monte Carlo simulations showed that the PDD is dependent on focal spot size.^{106,237} A smaller focal size beam gives a higher PDD for the same depth and field size than that of a larger focal spot beam, but the PDD remains almost unchanged without effect on focal size for fields ≥5 mm diameter. Spot size has not been shown to affect dosimetric parameters in many modern machines (manufactured after 2005) having a submillimeter spot size.²⁴⁶ The PDDs for a 5 mm diameter field, determined from measurements with diodes which have been corrected for dependencies and perturbations, were in good agreement (within 5%) with data from MC simulations.¹⁰⁶ In fact, the dosimetric protocols for photons have no correction factor for PDDs. TRS 483¹ and Francescon et al²³⁷ also confirmed that the effect of energy dependence is very small for small fields.

5.2 | Tissue Phantom Ratio (TPR)/Tissue Maximum Ratio (TMR)

TPR/TMR measurements are generally not required as input in advanced algorithms (kernel-based convolution/superposition, MC-based, or direct solvers of the Boltzmann transport equation). However, such data are useful for verifying calculations with these algorithms as well as to use as input in many MU check software tools.

TPR/TMR can be directly measured with the same detector systems as used for the direct measurement of PDDs.^{10,39,47,126,247,248} TPR/TMR measurements have an advantage for beam modeling in that there are no inverse square law effects to be considered. Direct measurement of TPR/TMR solves some additional problems related to PDD measurements: (a) the detector axis remains aligned with the beam axis at different depths because the detector is not moved; and (b) the ratio between the dimensions of the detector and the field dimensions is constant when the depth increases so the correction to readings remains constant, that is, $k_{Q_{clin}, f_{msr}}^{f_{clin}, f_{msr}}$ remains constant. Various approaches using standard water phantoms (such as PTW, Standard Imaging, IBA), innovative scanning systems (such as ARM Inc, Port Saint Lucie, FL), and direct measurement using automated couch movements of modern linear accelerator with 1-D tank²⁴⁹ can be used for the TPR/TMR measurements. The CyberKnife TPS system requires as input TMR data that can be obtained by mounting the detector in a birdcage fixed to the linac head and then moving the linac head along the z axis.

It has been shown that stereotactic diode and micro-chamber measured data are in good agreement with MC simulated data for very small fields (<2 cm × 2 cm).²³⁷ The simulation shows that the TMR measured with a stereotactic diode and a microchamber reproduces data within 1% of the TMR in water (excluding factors that cannot be considered by MC simulation such as dose rate dependence, polarity effect, etc.).

5.3 | Conversion between PDD and TPR/TMR

TMRs have been calculated for small fields from PDDs measured with stereotactic diodes and from measured phantom scatter factors.^{43,240,241,250–254} To determine TPR data for smaller fields (< 3 cm × 3 cm), Cheng et al²⁴⁰ fitted the measured TPR data with a third-order polynomial and extrapolated the zero-field TPR values. The fitted data deviated by <1.5% from MC calculated data for most depths. Xiao et al²⁵⁴ used a two-step fitting process on measured TPR data to derive a function whose results differ by about 1% from the measured TPR data. Yang and Pino²⁵⁵ proposed an analytical method for converting PDD measurements to TMR data for very narrow fields by parameterizing primary and scatter components. The calculated TMRs agreed within 2.0% with measured TMRs beyond the depth of maximum build-up. Several other methods,^{125,256–258} have been proposed to account for non-equilibrium conditions and to convert PDD measurements to TMR data within 1% in field sizes down to 0.5 cm diameter. The conversion formulas are based on equivalent field size. The scatter ratios assume full LCPE and the conversion requires S_{cp} (total scatter factor) and S_p

(phantom scatter factor) data for very small-field sizes. To avoid the issues related to the lack of LCPE, an empirical method was developed by Ding and Krauss.²⁵⁸ The empirical method can be used to obtain TPRs accurately from measured PDD curves for small fields down to a 4-mm diameter cone. Most of these methods are time consuming to apply in practice due to the analytical approach and have limited accuracy. It is recommended that any method adopted for small field TPR/TMR should be verified against the measurements, at least for a set of sampled points at various depths and field sizes.

5.4 | Profiles

The acquisition of beam profiles in small fields and the determination of the FWHM which specifies the f_{clin} parameter, needed to look up small-field output correction factors typically involve the measurements of line doses at orthogonal directions (gun–target and left–right) at a variety of depths and a variety of field sizes down to 0.5 cm × 0.5 cm. Much of the discussion in the previous sections also applies to the measurement of profiles. Primarily, it is recommended that the detector must be correctly mounted and precisely aligned (±0.1 mm) and oriented to the central axis of the beam at all depths. Use of a small-volume detector is extremely important for profile measurements to avoid incorrect determination of the width of the beam penumbra.^{5,240,242}

Figure 6 shows small-field profiles for a 6 MV beam and the effects of source size and field diameter. In this example, measurements with a diode detector are compared against the data calculated with the MC method. Good agreement is observed for fields of diameter as small as 5 mm with larger deviations seen for smaller fields. Volume averaging effects caused by detectors can greatly influence the quality of the measured dose profiles in terms of how well the detector can faithfully resolve the changes in the profile shape. Deconvolution methods that require knowledge about the effective area of the detector, which is modified by fluence perturbation^{5,259–262} can be used to correct the detector signal accordingly but it mitigates the volume effect alone. However, the energy dependence is also important and needs to be accounted. To account for energy response variation along a profile, Wegener and Sauer²⁶³ have proposed the convolution of the signals from 2 detectors of different types to resolve this issue in small field. A more accurate profile is shown for a smaller focal spot.¹⁰⁶ Therefore, small-field profiles should be measured with the smallest, least perturbing tissue equivalent detectors^{76,85,86,97,123,125,171,264} or radiochromic film.¹²³

Figure 6 adapted from Sham et al¹⁰⁶ compares MC data with diode-corrected data for relatively small fields. It also shows the effect of the focal spot size, with very

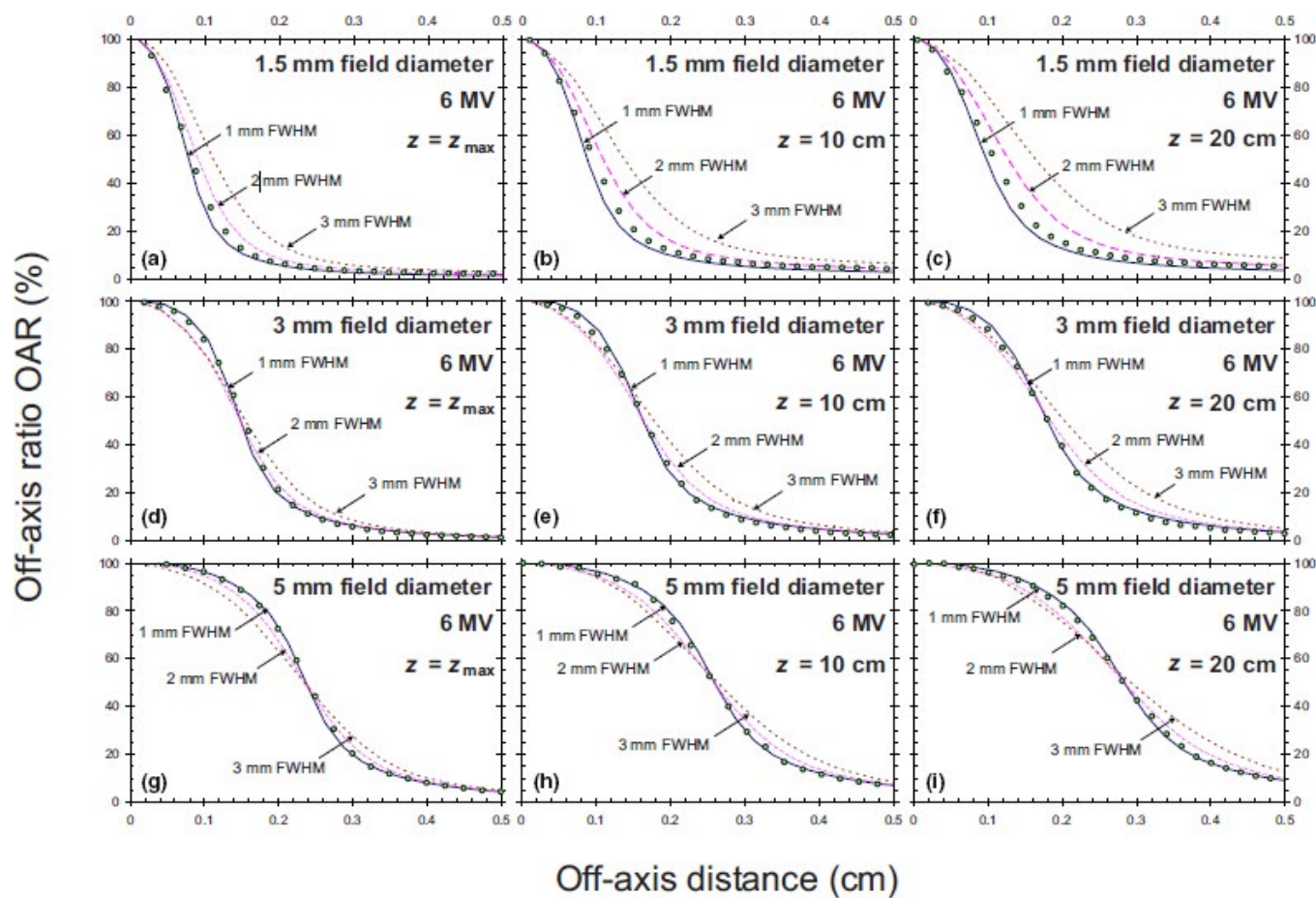


FIGURE 6 The influence of varying source sizes FWHM of a Gaussian focal spot distribution approximating the actual focal spot) on the off-axis ratio. Comparison of the calculation of small-field profiles with the Monte Carlo method with corrected diode data (symbol). (Reproduced from Sham et al¹⁰⁶ with permission) [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mp.15030)]

good agreement found between calculations and corrected measurements down to 5-mm diameter fields. The differences increase in smaller fields (Figure 6). MC simulations show that none of the detectors can exactly reproduce the profile due to perturbation by the detector materials in water. Additionally, Figure 7 shows profile (OAR) data from various detectors for a 6 MV beam for a 5 mm² field. When the perturbation factor is unknown, a detector with minimum perturbation should be used such as PSD (W2), microSilicon, and microdiamond.

Figure 8 shows the effect of detector materials on profile data.⁷² The differences in diode and microLion profiles are relatively higher only in the tail, where the doses are very low. Therefore, the profiles obtained with these two types of detectors can be considered a good approximation of the profiles in water. The data for the microchamber demonstrates poorer performance compared to the other two detectors, even in the penumbra region of the profile. When diodes are used, one ought to verify that the dose rate dependence does not change with distance from the central axis. When detectors with a directional asymmetry (e.g., ionization chambers) are used, they can be mounted vertically

(parallel to the beam axis), which can minimize the magnitude of penumbral blurring in both lateral and in-plane profiles. The user should first verify any significant stem effect or polarity effect. Additionally, the resolution of scanning in small fields should be kept to the limit of scanner motion which is typically 0.1 mm.

Radiochromic film may be used to compare with the detector-scanned profiles. This comparison requires adequate care to ensure careful film handling,^{20,190,191} energy, and temporal responses for the radiochromic film.^{265,266} The potential penumbral blurring associated with the film scanner may also need to be taken into account.

5.5 | Smoothing and data processing

When the PDD and profile data are acquired with the appropriate measurement equipment and a careful experimental setup, extended post-processing to the acquired data may not be necessary. Nevertheless, it may be reasonable to use a modest low-pass filtering technique (or smoothing) to remove any high frequency noise components (abrupt spikes, wiggles, etc.). Most beam data acquisition system

FIGURE 7 Beam profiles showing the OAR of the 5-mm collimator measured at a depth of 10 cm with the Exradin W1 PSD scintillator detector, PTW 60012 diode, and PTW 31014 microchamber with the stem axis parallel to the beam axis before and after applying the correction factor for 5-mm collimator from CyberKnife. Adapted from Francescon et al⁷² [Color figure can be viewed at wileyonlinelibrary.com]

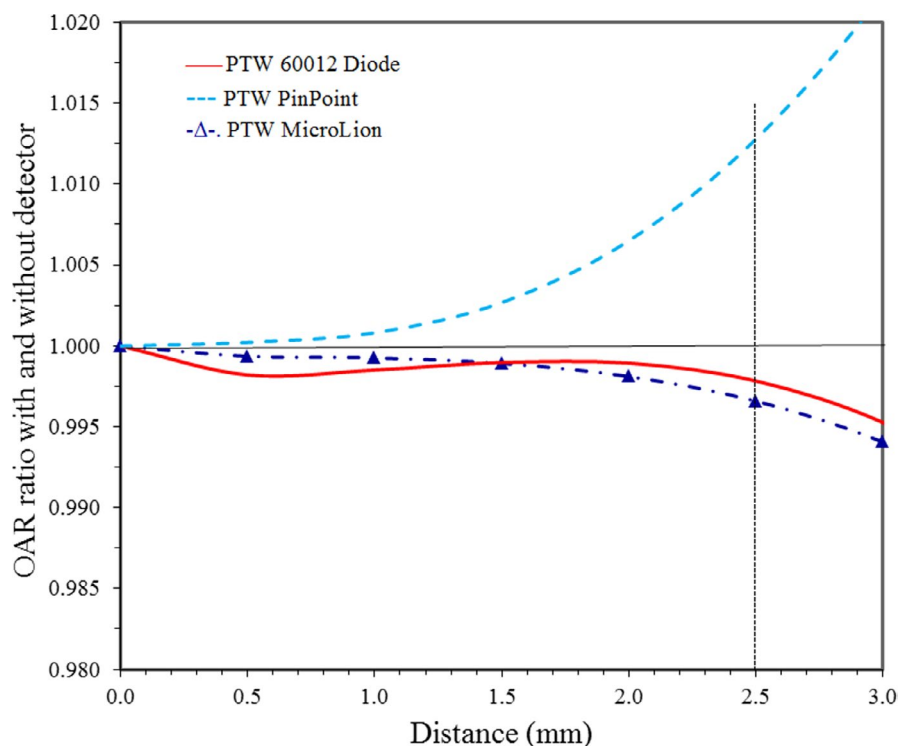
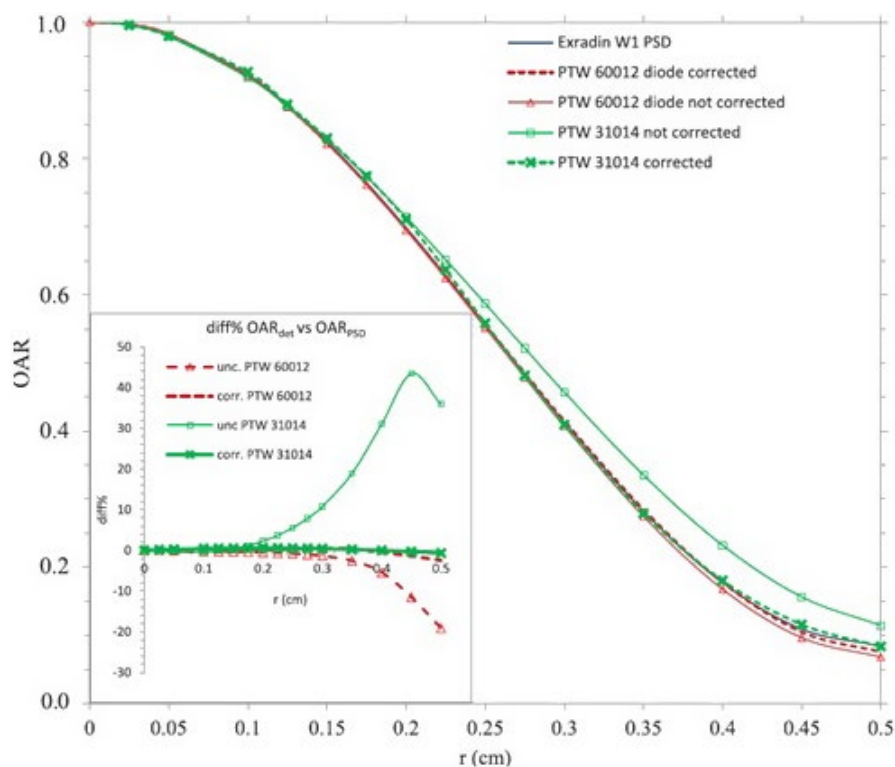


FIGURE 8 Monte Carlo simulated ratios of the off-axis ratio (OAR) with and without consideration of detector in water for the 5x5 mm² field for 6 MV beam (modified from Francescon et al⁷²). The detector configuration is explicitly included in the simulation to obtain the most accurate results when compared with experimental profiles. For diode and microLion detectors, the differences are relatively higher only in the tail region. Therefore, the profiles obtained with these two detectors can be considered a good approximation to the profiles in water. The PinPoint detector on the other hand shows worse results for the profile than the other two detectors, even in the penumbra region of the profile, due to its characteristics including the material composition. The field edge is indicated by the line at 2.5 mm [Color figure can be viewed at wileyonlinelibrary.com]

software provides a number of data smoothing options using a wide range of algorithms. Regardless of the filtering/smoothing technique that is used, it is imperative that the original dataset is retained to ensure that there are no unintended changes to the data such as artificial penumbral blurring or altering the peak of the profile. These considerations are especially important for scanned data for narrow field sizes. Typically, median filters are well suited for smoothing profile data, as the original data points are preserved. Median filtering should be applied to well-sampled profiles (i.e., with high point density or good resolution (0.1 mm)). The restrictions on PDD smoothing are less critical, and median or least-square fitting or other approaches may be sufficient. There is no general rule for how much smoothing can be applied to the data. Any post-processing to the data should not distort dose gradients present in the profiles, namely doses in the build-up region and/or the penumbra region. The originally scanned, raw data must always be saved as reference.

5.6 | Field output correction factors

A sample of representative field output correction factors for various detectors is shown in Figure 9. More comprehensive tabulated data for wide range of detectors can be acquired from TRS-483¹ and from recent publications. Figure 9 demonstrates the wide variation of $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ correction factors for different detectors for small fields.

Figure 10a shows the difference of $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ from 1.0 plotted on the y-axis versus various types of detector for beams from a Varian accelerator (Das et al)¹²⁴ Similar data for CyberKnife is shown in Figure 10b. As one can see, there are certain detectors where the difference is relatively large and should not be used. Therefore, only detectors such as plastic scintillator, microdiamond, and microLion are suitable due to their smaller $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ values. As shown by Liu et al 2014,²⁶⁷ the measured small-field data for a 6 MV beam is relatively

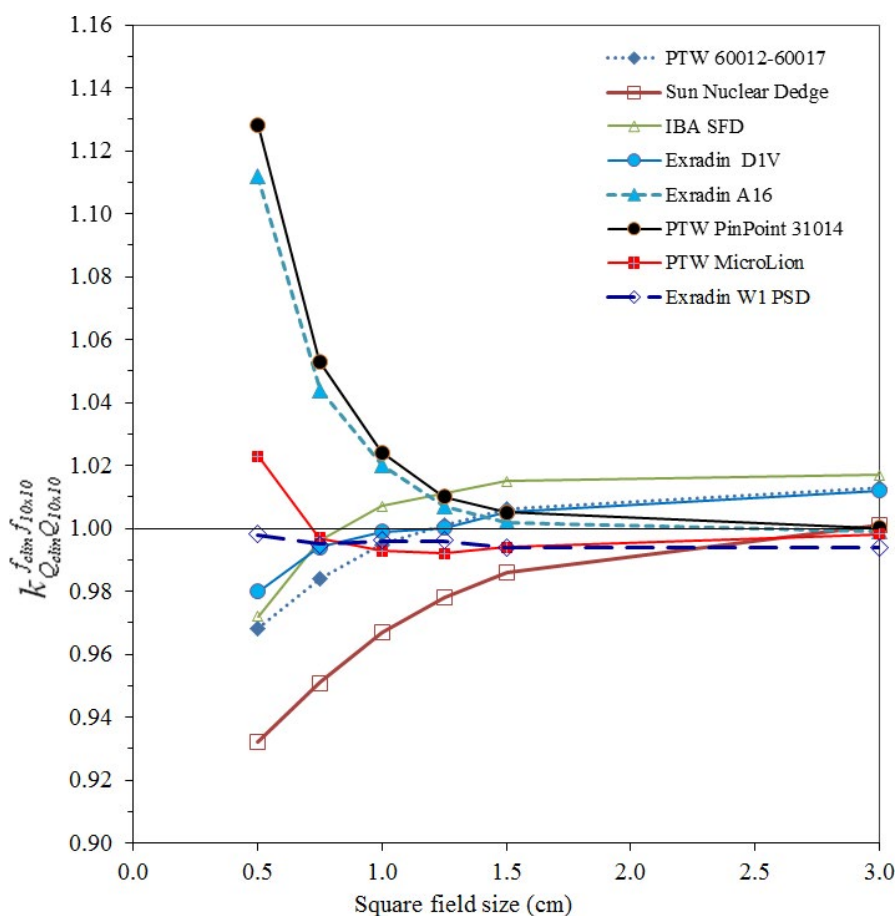
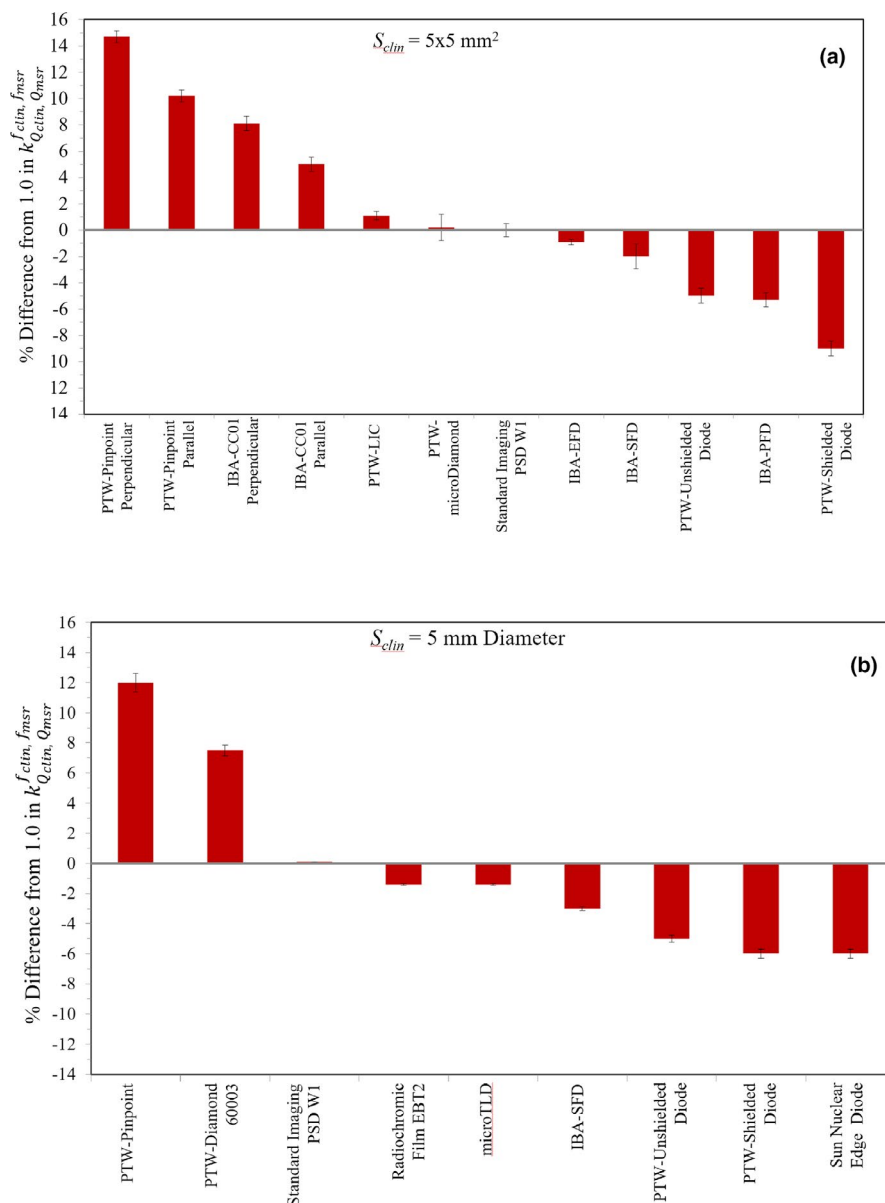


FIGURE 9 Effect of field size on the detector-specific correction factors. $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ was calculated as the mean over all the values obtained by changing the linac models, (Siemens Primus and Elekta Synergy 6 MV), the radial FWHM and energy of the electron source as presented by Francescon et al.⁷² These values can also be used with caution for Varian 6 MV machines. The maximum statistical uncertainty is ± 0.003 (1 σ). Data from TRS-483¹ for various detectors at 10 cm depth differed for small fields. The differences between studies could be due to variations in depth (d_{max} vs 10 cm) in the Monte Carlo simulations [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

FIGURE 10 Variation in the detector-specific correction factor induced by different detectors. (a), The variation of $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ from unity for 5x5 mm² field is plotted for a wide range of detectors for a Varian IX machine and (b) for a CyberKnife unit with a 5 mm diameter cone. Adapted from Das et al.¹²⁴ [Color figure can be viewed at wileyonlinelibrary.com]



insensitive of the machine manufacturer for a given detector.

Even among the best detectors, there is a large variation in the correction factors as shown in Figure 11 when the field sizes are $<1 \times 1 \text{ cm}^2$. For fields $<1 \times 1 \text{ cm}^2$, accurate measurement results require that an appropriate correction factor is chosen. For fields $\geq 1 \times 1 \text{ cm}^2$, the correction factor would be smaller in magnitude making it more straightforward to accurately measure the dose as long as a proper detector is chosen. This is reflected in 0.8% variation for $1 \times 1 \text{ cm}^2$ fields for most detectors as shown in caption.

6 | DOSE MODELING IN TREATMENT PLANNING

Treatment planning system dose calculations represent a critical link in the chain of clinical dosimetry events

that determines the accuracy of the dose given to a patient. There are several guidance documents providing recommendations regarding commissioning and verification of TPS dose calculations.²⁶⁸⁻²⁷⁰ TG-105²⁷¹ provided a detailed discussion on treatment planning based on MC simulation, but it did not go into detail on small-field modeling or dosimetry. ICRU Report 91⁹ provides some basic principles relevant to TPS performance in small fields; however, there is as yet no dedicated recommendation focusing on small field TPS commissioning and verification. The goal of this section is to briefly summarize a number of important elements implicated in the commissioning of a TPS for dose calculations in small fields. It does not address the clinical process from simulation to treatment planning and delivery for small fields.

To accurately model the fluence for a small field, all causes of gradients need to be accurately modeled,

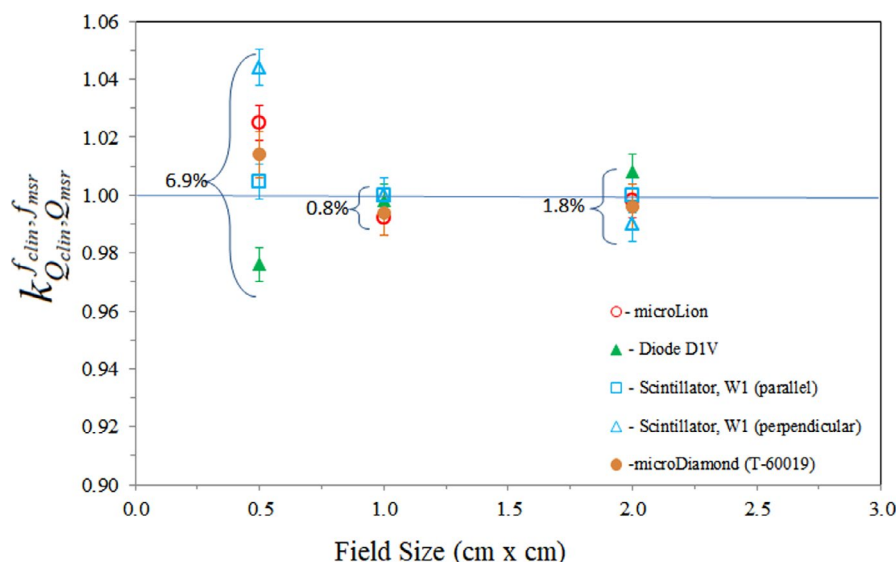


FIGURE 11 Variation of $k_{Q_{clin}, Q_{msr}}^{f_{clin}, f_{msr}}$ versus field size for four suitable detectors (courtesy of Indra J. Das). Please see that the variation is larger for smaller fields [Color figure can be viewed at wileyonlinelibrary.com]

including source occlusion, collimator leaf tip leakage, and collimator setting corrections. The modeling freedom for clinical physicists is often limited because TPSs commonly require the same set of beam model parameters for all field sizes. A workaround to mitigate insufficient modeling is to use different sets of parameters for different field size ranges in order to minimize resulting dose errors. Multiple field size ranges lead, however, to challenges in the clinical environment with respect to limiting choices, issues with plans that include both small and medium-sized fields, and the risk for using an incorrect beam model for a given patient's treatment plan.

The smaller a limiting field size is set, the more critical is the consistency and interplay between beam modeling, delivery, and measurements since errors in any of these may cause discrepancies. This approach can be further complicated by competing needs for SRS and SBRT programs and IMRT and VMAT plans. It is clear that tools for handling and verification of small-field modeling are critical for TPS functionality.

6.1 | Treatment planning dose calculations

Dose calculation algorithms used for treatment planning are generally classified as factor-based algorithms or model-based algorithms. According to ICRU Report 91,⁹ factor-based models are inappropriate for use with very small fields when the irradiated medium deviates significantly from the homogeneous water medium in which the input basic dosimetric data are measured. Since this represents the majority of the important clinical applications, the remainder of section VIII will only address model-based calculations. Model-based dose calculations involve two separate steps: modeling based on field settings gives the energy fluence exiting the treatment machine, and modeling of the dose

deposition resulting from irradiating a patient/phantom with that fluence. The energy fluence modeling is independent of the patient geometry and can thus be independently verified. A normalization versus a calibration measurement of absolute dose per monitor unit for a reference geometry is commonly done to yield dose per monitor unit from the TPS. The energy fluence per MU in air in the beam's isocenter plane can be viewed as composed of fluence of the direct source and fluence from different scatter sources such as flattening filter and collimators normalized to the monitor signal consisting of particles moving forward as well as those that are backscattered. For small fields, the scatter distributions from the flattening filter and collimators are relatively small without large variations so the major contributor to the overall energy fluence is the direct source. Hence, the size and shape of the direct beam source, collimator geometry, and the transmission through the MLC leaf tips are the most critical phenomena in modeling small-field fluence for subsequent accurate dose calculation. This can become quite involved and various approximations are often used to simplify implementation or speed up calculations. Unfortunately, the applicability range of approximations is not always well known, challenging the clinical physicists for verification.

6.2 | Direct beam source size and shape

Occlusion and partial attenuation of the primary source by the collimator leaf ends are the single most important processes that drive the small-field characteristics of a linac. Most TPSs represent the primary source by a 2D Gaussian, occasionally with different widths in the in-plane and cross-plane directions. Even a point monodirectional electron beam is likely to result in a Gaussian-shaped photon source model due

to multiple scattering in the target. Direct experimental determination of the size and shape of the direct source^{100,101,103,107,272–274} reveals that it indeed varies with linac design and/or with energy on the same linac. Some bending magnet arrangements tend to yield prolonged, elliptical source shape profiles. The FWHM can vary over several mm in range, and eccentricities up to 3 mm (Figure 3).^{100,101} Furthermore, the shape of the focal spot for fields smaller than 2 cm × 2 cm also affects the shape of the depth dose curves.^{69,106,229} Measurements of source sizes are not part of common clinical routine, although their critical role for small-field output modeling deserves more attention. The source size is often used as a fitting parameter during commissioning of TPS algorithms by fitting calculated dose profiles for large fields to their measured counterpart. Small-field output factors are particularly sensitive to source size settings because of the overlap of penumbral influences from any nearby field borders. While beam fitting based on small fields provides more sensitivity for source modeling, this method is rarely recommended by TPS vendors to their users due to uncertainties in the small-field measurements. However, carefully measured small-field output factors should be used for verification of the field sizes considered for clinical use.^{69,106,229}

6.3 | Collimator geometry and fluence modeling

After source occlusion, beam shaping is affected by the leakage profile through the rounded ends of MLC leaves (when present). In principle, the fluence modeling needs to trace individual rays from different parts of the source all through the MLC leaf ends to all positions in a beam exit plane as to ensure full accuracy of the result. Some TPSs do that, but approximations are common. Modeling the leaf attenuation profile as a shift in collimator position proportional to the effective gap offset defined by Losasso et al²⁷⁵ (in later literature called the dosimetric leaf gap (DLG)²⁷⁶) has been shown to be insufficient.²⁷⁷ A workaround for such TPSs has been suggested²⁷⁷ through use of values optimized to minimize deviation over ranges of field sizes rather than use the directly measured dosimetric gap. The measured DLG for IMRT is unidirectional (either left to right or right to left), whereas it is bidirectional for VMAT which imposes challenges in beam modeling.^{278,279} More accurate modeling can be achieved by combining explicit leaf tip projection corrections with a modulated transition region which may be beneficial for clinical use.²⁸⁰ Using the IROC-Houston head phantom, Brandon et al²⁷⁹ provided data on the error in treatment planning commissioning associated with the MLC offset setting. It was noted that for the PTV the coverage varied from –2%

to +10% and for spinal cord –12% to +20% for values ranging from –2 mm to +2 mm of the MLC offset, respectively. It is also observed that there is a strong dependence on the PTV size with the largest differences for higher DLG.²⁷⁸

It follows that validation of the TPS interpretation of the collimator geometry is pertinent⁵¹ but may not be sufficient due to the approximations. If several layers of collimators are aligned, the resulting source occlusion will be sensitive also to alignment errors. Thus, accelerator-operating modes that retract the backup collimators from the field edges are preferable to ensure robust collimating conditions. The broadening of the fluence penumbra due to source occlusion will be larger when the collimator is higher up in the beam path. Collimator positioning is critical for verification. Central axis position offsets^{281,282} should be considered and checked as these normally are included in the linac software. However, for small fields, aperture sizes estimated from FWHM may not map linearly to collimator settings due to non-equilibrium conditions that may cause bias leading to field size overestimation.^{275,283}

6.4 | Energy fluence mapping and verification

During the commissioning process, it is helpful to review fluence maps with an adequate resolution (e.g., 0.1–0.2 cm) for a qualitative evaluation of the modeling of penumbra details, tongue-and-groove effects, and interleaf leakage. The energy fluence should be given in a normalization mode which is fully linkable to head scatter factors to facilitate verification.⁶⁴ For patient plans, viewing the fluence map may be helpful when IMRT or VMAT is used for MLC-based delivery systems. The complexity of a field can then be analyzed as a precursor to measurement-based QA for patients' treatment with VMAT to multiple spatially separated targets.^{284,285} This information can be used to determine if a plan may need to be modified prior to measurements. This has value because the patient population often requires to be treated as quickly as possible and delays due to failed measurements would hinder patient care.

6.5 | Dose calculation considerations

For small fields, model approximations play an important role when the dose calculation engine of a TPS calculates the resulting dose distribution from an energy fluence distribution.²⁸⁶ With increasing beam energy, the lateral range of secondary electrons also increases. At higher beam energies (>6 MV), electron transport dominates the penumbra shaping, while at energies around 6 MV, both electron transport and source occlusion contribute to the penumbra shaping.^{220,287}

Detailed descriptions of different algorithms and their expected performance in different situations are given in several references.^{116,277,280,288} Algorithms such as pencil beam convolution (PBC) are not suitable in small fields due to many factors especially the poor lateral electron transport.²⁸⁹ Even in the larger fields used in SRS/SBRT, dosimetric discrepancies between PBC and collapsed cone (CC) algorithms have been shown to be as high as 16%.^{290,291} Numerous investigations have demonstrated its applicability to calculate dose under disequilibrium conditions, provided it is correctly configured, but statistical noise is associated with its use. Because of a large amount of lung in the field, the thorax region is challenging and presents difficulty in dose calculations due to the lateral disequilibrium where PBC approaches fail and are inadequate. Point kernel superposition/convolution models perform well for lung densities (with the exception of small deviations close to interface regions) if the kernels are correctly scaled to take into account lateral electron transport due to locally varying heterogeneities. Models intermediate between pencil and point based kernel also exist (e.g., the AAA model which is a pencil based model that includes an approximation for lateral electron scattering). In summary, for homogeneous, near-water-like tissue densities, a PBC model's final accuracy mostly depends on the accuracy of the beam model.^{69,99,111,277,280,286,292} However, in heterogeneous calculations, PBC is not recommended in small-field applications. For these reasons, MC is often used to benchmark the implementation of different algorithms^{114,115,292,293} and in some TPS systems, it is also directly used for patient dose calculations. MC dose calculations explicitly account for fully coupled photon and electron transport.

Besides approximations of the physical processes involved in dose deposition, discretization effects depend on the numerical implementation in the TPS algorithm and will influence the final dose accuracy. The common collapsed cone approximation for speeding up point kernel computations discretizes the directions to a limited set in which the radiant energy is spread from a primary point of interaction.^{286,294} Although the electron transport significantly blurs the dose deposition pattern, resolving small targets and organs-at-risk may still require a high resolution for the dose calculation. In particular, for non-MC algorithms an isotropic grid of 2 mm is recommended while for MC algorithms, voxel sizes between 1 and 2 mm on a side are considered appropriate (AAPM TG-101,⁸ AAPM TG-105,²⁶⁵ ICRU Report 91⁹).

6.6 | TPS configuration and verification for small-field modeling

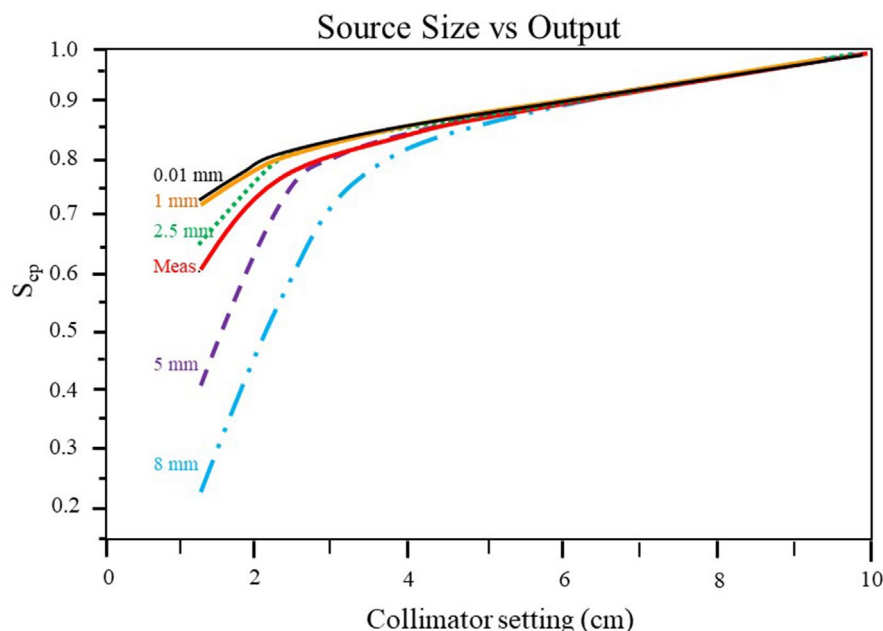
The configuration of the fluence and dose engines for model-based TPS algorithms, that is, convolution/superposition or MC simulations, requires the linac

head geometry data as well as a set of measured dosimetric data (depth functions, profiles and output factors). The data are commonly used in an iterative process to set certain beam model parameters, such as the energy spectrum, the shape of the primary energy fluence distribution, and the width of the focal spot.^{64,295–298} Errors in the measured data can therefore lead to an inappropriate choice of modeling parameters implemented in the TPS. The choice of the focal spot size in particular is of importance for small fields.^{69,106,126} A value for this focal spot size may lead to the TPS performing optimally for broad beams, but less well for narrower beams. Source size model also has an impact on the penumbra of larger fields (e.g., 5 cm × 5 cm) that many physicist's tune to fit the data. Figure 12 shows how the calculated output ratio can vary with source size in a beam model and also illustrates how measurements with an inappropriate detector can affect the beam modeling. In the case of deviations, TPS model parameters may need to be adjusted appropriately. Some vendors have recommended instead that a separate beam model (treatment machine) be defined in the TPS for use with small fields (usually collimated to less than 1 cm × 1 cm). This strategy can be an impractical work around for IMRT planning if optimization engines result in a mixture of beam segments with small and large sizes.

In addition to the standard practice of verifying the modeling of small fields for homogeneous water medium, a model should preferably be tested in inhomogeneities, in particular for low-density materials.^{269,299} While making such measurements in inhomogeneous materials are delicate, point dose measurements are worthwhile to verify inhomogeneity corrections, as performed by Tang et al³⁰⁰ and Saxena et al³⁰¹ Evaluation of calculations for inhomogeneous media is important and recommended by multiple AAPM guidance documents^{13,299,302} These reports as well as guidance from ASTRO^{303,304} also stress the value of testing a system from the beginning (often CT simulation) to the end (treatment delivery) which is also referred to as an end-to-end test. There are several commercial phantoms that include heterogeneous media that can be used with film or OSLDs. In addition, an independent evaluation such as available from IROC-Houston (<http://irochouston.mdanderson.org/RPC/home.htm>) provides invaluable information on the accuracy of the dose calculation and the entire treatment process.^{305,306}

Published results that are custom to small fields use low-density media, such as lung-equivalent plastic, Styrofoam, cork, or balsa wood could be used as an example.^{292,307,308} As some commercial TPSs made MC algorithms available, these low-density media should first be investigated for conventional field sizes to validate the beam model.^{114,115,292,309–313} As noted above, either a commercial phantom could be used or a

FIGURE 12 Illustration of variation in the calculated output factor with different effective source diameters used in the beam modeling of Helax-TMS treatment planning system. A diameter of 2.5 mm, which gave the best match in field output factors at $d_{\max}$ (S_{cp}) between measurement and calculation for collimator settings greater than 3.0 cm, was used to configure the TPS for use with broader beam sizes greater than 3.0 cm. Adapted from Aspradakis et al³²⁸ with permission [Color figure can be viewed at wileyonlinelibrary.com]



phantom-by-mail service such as from IROC-Houston. When a TPS will be used in small photon fields, the characterization and modeling of the beam sources and collimating devices should be particularly checked for the following:

1. Source size setting by verification of small-field output for central and off-axis fields (see, e.g., Figure 12);
2. MLC interleaf leakage and leaf-side modeling
3. Jaw and leaf-end field edge positioning and transmission at different off-axis positions
4. Backup jaw and MLC leaf-edge interaction (position of jaw with respect to MLC)
5. Collimator intraleaf leakage modeling
6. Penumbra, and full profile when penumbra overlap

In low-density media, widening of penumbra, build-down, and re-build-up for beam passages through low-density regions are important considerations. A difficulty lies in validation of IMRT fields or VMAT arcs, which are composed of many small fields in this media. While most models may work well for IMRT if the MLC parameters (such as transmission, tongue, and groove, etc.) are accurately modeled, the same algorithm may fail or be less optimal for VMAT treatment plans.³¹⁴

7 | UNCERTAINTY

This section deals with uncertainty in field output and profiles. A detailed analysis of type A and B uncertainties for small-field measurements is provided in TRS-483.¹ It is been shown that measurement uncertainty increases from $\pm 0.2\%$ for fields $>10 \text{ cm} \times 10 \text{ mm}^2$

to 3.3% for $5 \text{ mm} \times 5 \text{ mm}$. An uncertainty analysis framework has been universally adopted from the Joint Committee.³¹⁵ Sources of these uncertainties include:

1. Misalignment of the detector axis with respect to the beam central axis
2. Deviation of the vertical motion of the scan arm from the central axis
3. Inaccuracy in the collimator jaw settings
4. Spectral changes with field size resulting in incorrect mass stopping-power ratios, especially for linear accelerators operating without a flattening filter
5. Incorrect or unknown correction factors to convert detector signal to dose
6. Incorrect SSD or measurement depth.

A detailed analysis of the uncertainties related to the relative output measurements of small beams is reported in various references^{66,67,70,72,89,316,317} and has been estimated to be about 0.7% at one standard deviation (SD) in a small field in a homogeneous medium. The analysis that considers both type A and type B uncertainties on the correction factor $k_{Q_{\text{clin}}, Q_{\text{msr}}}$ indicates that this quantity can be provided by MC simulations with an overall uncertainty lower than 0.7% (1 SD).⁷² Type A uncertainties are obtained through statistical methods. Conversely, type B uncertainties are related to non-statistical quantities such as errors in the underlying cross-section data and geometry approximations used in modeling the detector which are not influenced by the user but are obtained through other methods. From these considerations, the field factor defined in Equation 3, has an associated overall uncertainty that depends on field size and detector but is about 1% (1 SD),

considering the contribution of uncertainties of output measurements and $k_{O_{clin}, O_{msr}}^{f_{clin}, f_{msr}}$ by MC calculation. However, this is dependent on the detector and the field size. TRS-483¹ has provided detailed analysis and data for various detectors that varies from 0.7% to 3%.

The uncertainty of the profile measurements can be estimated by using the GUM (Guide to the expression of Uncertainty in Measurements) report.³¹⁵ The contributing uncertainties that should be considered are:

1. Source-surface distance within 1.0 mm
2. Horizontal position of the phantom within 0.2 degree
3. Depth of the detector within 0.15 mm
4. Position of the detector in the x, y directions within 0.1 mm is clinically acceptable, even though uncertainties of 0.05 mm and 0.01 mm are reported in the literature
5. Reproducibility of the detector response is truly achievable at the <1% level or better at least in central region of a profile.^{20,89,317} With modern detectors and electrometers, where conditions can leverage with a high signal to noise ratio (nC to pC, giving 1000 folds advantage, e.g., 0.1%), this can even be achieved within 0.1%–0.2%.

For the profile the uncertainty is variable as the detector moves along the axis. It increases and has a maximum value where the gradient of the dose profile is maximum. A total uncertainty on $OF_{det}^{f_{clin}}$ measurements for the smallest field (0.5 cm × 0.5 cm) was found to be 0.7% at 1 standard deviation as reported

by Francescon et al.⁷² The uncertainties calculated by separating the contributions to the uncertainty into the three main components, that is, detector positioning, field size reproducibility due to the mechanical tolerances of MLC and output fluctuations of the accelerator were 0.2, 0.6, and 0.4%, respectively, and when summed in quadrature gave a total relative uncertainty on $OF_{det}^{f_{clin}}$ of 0.75% (1 σ). The two results, obtained with two different procedures, the first one providing directly the total uncertainty by measurements and the second one calculating the contributions of the different sources of uncertainties, provide a very similar and consistent result. By the analysis of the three separate uncertainties which contribute to the total uncertainty on $OF_{det}^{f_{clin}}$, we can see that the main source of uncertainty is due to the reproducibility of the field size due to the mechanical tolerance of MLC, assuming the uncertainty due to detector positioning is negligible due to high precision of most water tank systems.

The estimation of MC uncertainties (type A and type B) of $OF_{det}^{f_{clin}}$ and of $k_{Q_{clin}, Q_{10 \times 10}}^{f_{clin}, f_{10 \times 10}}$ for several detectors and for the smallest field (0.5 cm × 0.5 cm) is also reported.⁷² For larger fields, the geometrical uncertainty is expected to be lower and comparable to the values calculated by Wulff et al.³¹⁸ The total uncertainty (types A and B) due to the MC calculations of $OF_{det}^{f_{clin}}$ and of $k_{Q_{clin}, Q_{10 \times 10}}^{f_{clin}, f_{10 \times 10}}$ is less than 0.7%.^{89,317}

The uncertainties can be composed in a geometric mode as discussed in GUM³¹⁵ by using the MC procedure as proposed in several references^{72,88,318–320} and shown in Figure 13. The uncertainties, largely dominated

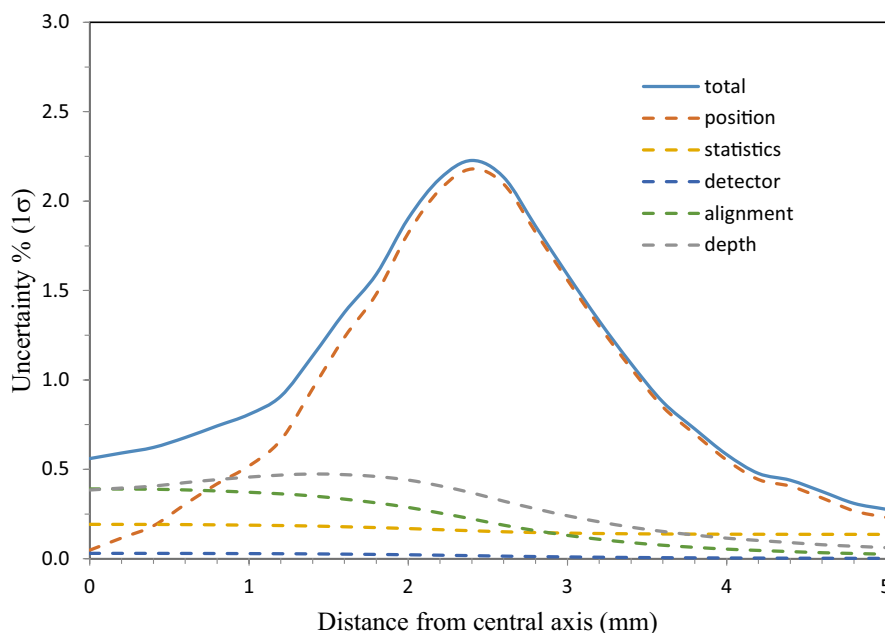


FIGURE 13 Fractional uncertainties (1 σ) for a dose profile at 10 cm depth of a 5x5 mm² nominal field size of an Elekta Synergy 6 MV machine along the y direction (lower diaphragm). The uncertainties considered are: Total =total combined uncertainty, Depth =depth of the detector, Position =position of the detector in the y direction, Alignment =misalignment of the detector axis with respect to the beam central axis. Statistics = fluctuations of the machine output, Detector = sensitivity of the detector. Conservatively, the statistical distribution is considered uniform (Figure courtesy of Paolo Francescon) [Color figure can be viewed at wileyonlinelibrary.com]

by the dose gradient and the detector sensitive volume. Additionally, it is usually greater for the y profile (lower jaws) compared to the x profile collimated by the upper jaws.

8 | KEY RECOMMENDATIONS AND SUMMARY

These recommendations apply to small fields. A small field is defined as one in which lateral charged particle equilibrium cannot be established (see Table 2). The definition of a small field depends on the beam energy. In general, fields $\leq 3 \text{ cm} \times 3 \text{ cm}$ can be treated as small for most photon energies (Section 3). It is reported that a few microchambers can be used without correction factor on the determination of output factors for fields $>1.5 \text{ cm} \times 1.5 \text{ cm}$ or diameter $>1.5 \text{ cm}$.¹⁸⁵ Such detectors can be used both with the stem axis parallel or perpendicular to the beam axis. For fields $<1.5 \text{ cm}^2$, it is recommended that these are placed with the stem parallel to the beam axis with specific correction factors to convert the reading to dose as shown in TRS-483.¹

8.1 | Detector choice considerations

- For fields less than $1 \text{ cm} \times 1 \text{ cm}$, the interplay of source occlusion and detector dimensions demands careful selection of the detector (Section 6.6, Figure 12).
- Select a small-volume detector with minimal energy, dose, and dose rate dependence (Section 4).
- A detector with a known $k_{Q_{\text{clin}}, Q_{\text{msr}}}^{f_{\text{clin}}, f_{\text{msr}}}$ factor that is preferably close to one¹²⁴ should be selected, and proper measurement methods should be used (Section 6.6). The values of $k_{Q_{\text{clin}}, Q_{\text{msr}}}^{f_{\text{clin}}, f_{\text{msr}}}$ for various detectors are provided in the TRS-483 that could be used unless a revised value from recent literature is available as shown in reference for microDiamond.^{124,176,177}
- Unshielded stereotactic and electron diodes, plastic scintillators, microdiamonds and microSilicon are suitable detectors for measurements in small fields (Section 5.6, Figure 10).
- The advantages of plastic scintillator detectors (W1 and W2) are their small size, tissue equivalence, angular independence, and density similar to water.
 - The W1 plastic scintillator may have difficulty discriminating spectral sensitivity in small fields (Section 4.1.4).
- For field sizes $\leq 1 \text{ cm} \times 1 \text{ cm}$ (Section 4.1.2), an unshielded stereotactic diode, electron diode, plastic scintillator, and single crystal microdiamond are recommended.
- For field sizes $>1 \text{ cm} \times 1 \text{ cm}$, very small ionization chambers (micro-ionization chambers)⁵ (excluding those with high-Z electrode) are suitable.
 - The response of an ion chamber is less affected by variations in low-energy photon fluence than diodes.
 - The selected detector should be evaluated for the signal-to-noise ratio and the measurements corrected for any polarity effects³²¹ and volume averaging (Section 4.1.1.2).
- To use radiochromic film as one of the dosimeters, a flatbed scanner should be commissioned for dosimetry applications and a robust program with respect to the handling and processing of the film as described in TG-235¹⁹² (Section 4.2.1).

8.2 | Measurement considerations

- For fields less than $1 \text{ cm} \times 1 \text{ cm}$, the interplay of source occlusion and detector choice demands precise positioning of the detector.
- The resolution of the scanning measurement should be smallest possible to the limit of device usually 0.1 mm .
- Dosimetric measurements should be performed with more than one detector system.
- Strategies to minimize variations in measurements among users should be tested, utilized, and shared
- The daisy chain approach^{148-150,322} can be used to determine output factors for the complete range of field sizes needed clinically. However, it should be remembered that this method normalizes the response increase in large fields, it does not account for the fluence perturbation effect that may require a correction. It is emphasized that if better detectors are available, they should be purchased and used so one does not need to over-rely on the daisy-chain method (Section 4.1.2).
 - In this situation, unshielded stereotactic diodes can be used to measure the output in field sizes $\leq 3 \text{ cm} \times 3 \text{ cm}$ and small ionization chambers be used to measure the output from $3 \text{ cm} \times 3 \text{ cm}$ to $10 \text{ cm} \times 10 \text{ cm}$ reference field.
- While the changes in energy spectrum may be significant for small fields, the effect on the changes in mass stopping-power ratios is negligible ($<0.5\%$).^{36,145,323,324} Hence, the actual factors that contribute to $k_{Q_{\text{clin}}, Q_{\text{msr}}}^{f_{\text{clin}}, f_{\text{msr}}}$ are the various perturbation factors as shown in Eq.5. (Section 2.2)
- The electrometer and triaxial cables as well as any connector and additional cables need to be of a high quality to measure signals in the 10^{-12} – 10^{-13} C range. The effect of leakage current should be evaluated and then corrected in the measurement when necessary (Section 4.1.1.1).

8.3 | TPS commissioning considerations

- Measured data that are required as input for TPS configuration and beam modeling need to strictly

adhere to the requirements of the TPS manufacturer (Section 6).

2. Measured data should be compared with specific published data from a similar machine. If a difference between published and measured data is observed, the sources of the discrepancies must be understood in the context of source size, measurement and uncertainties, as described.^{100,102-104,274} Significant discrepancies should be resolved prior to clinical use (Section 6).
3. The TPS performance should be validated with corrected measurements from two recommended detectors when used for treatment planning with small fields.
4. Although pencil beam dose engines are expected to perform reasonably well in homogeneous media, their use in inhomogeneous medium is not recommended. Dose engines based on the MC method, Boltzmann equation solvers, or point kernel convolution/superposition are the most accurate method for modeling dose from small fields within inhomogeneous media.
5. A fine calculation grid size (such as ≤ 1 mm) should be used for small-field applications (Section 6.6).

8.4 | Independent checks

1. Small field dosimetric measurements should be verified independently, either through measurements or through an independent external audit, such as that carried out by IROC-Houston^{325,326} or in European countries with postal dosimetry services as described by del Mar Espinosa et al³²⁷ referring to published data would also be helpful.
2. A phantom test from CT simulation through treatment delivery, known as an end-to-end test, should be performed and may involve an independent assessment of the delivered dose (Section 6.6).

As noted in the introduction, there are a number of guidelines and recommendations related to small-field dosimetry. This report focused on relative dose measurements with respect to detector choice and the importance of more than one appropriate detector for these measurements. Important considerations for the measurement setup were defined as well as some limited information regarding TPS calculations for small fields.

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CONFLICT OF INTEREST

The members of [TG-155] listed below attest that they have no potential Conflicts of Interest related to the subject matter or materials presented in this document.

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